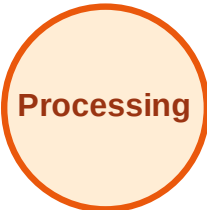


BFS Traversal

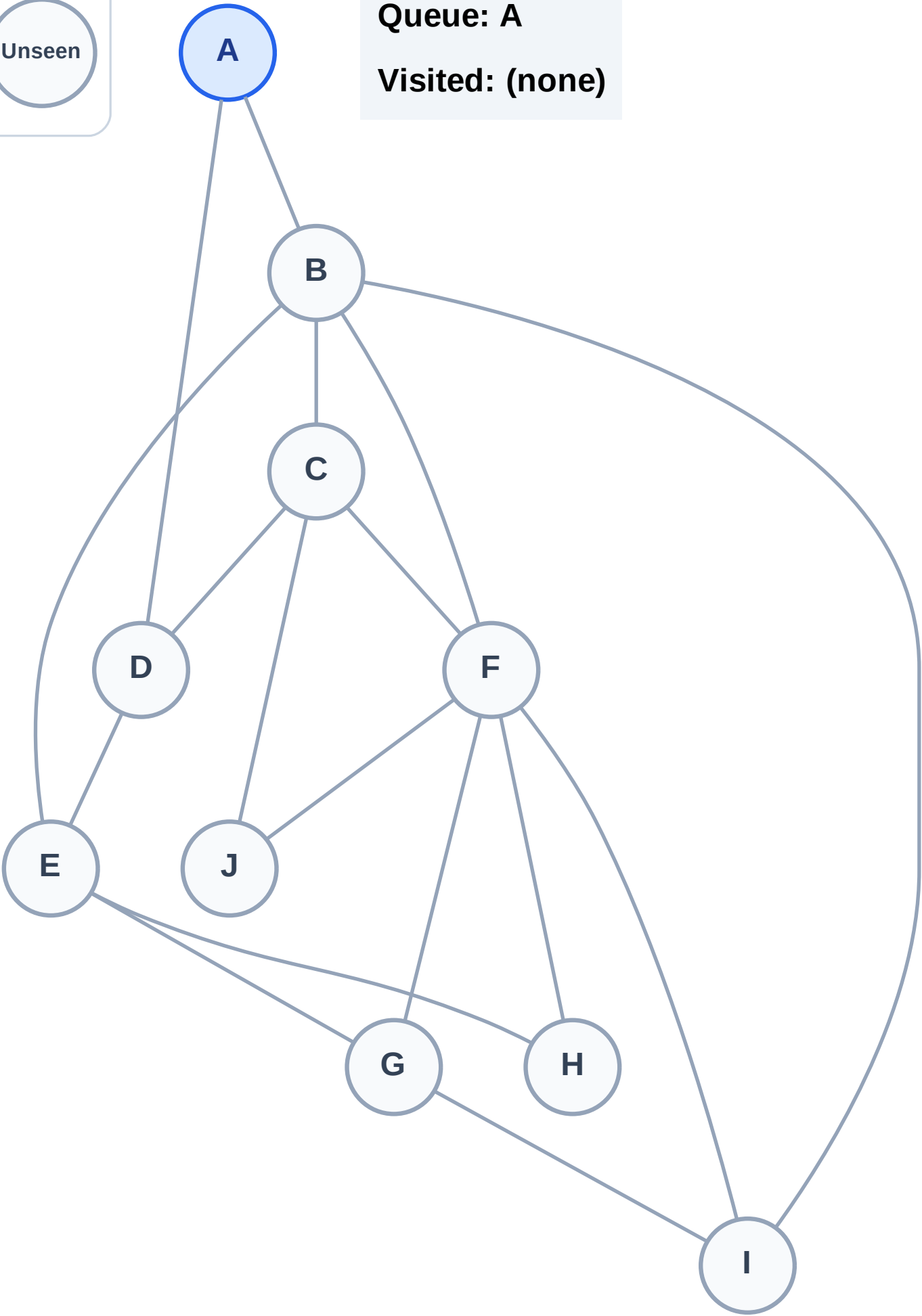
Step 1: Start at A

Legend



Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

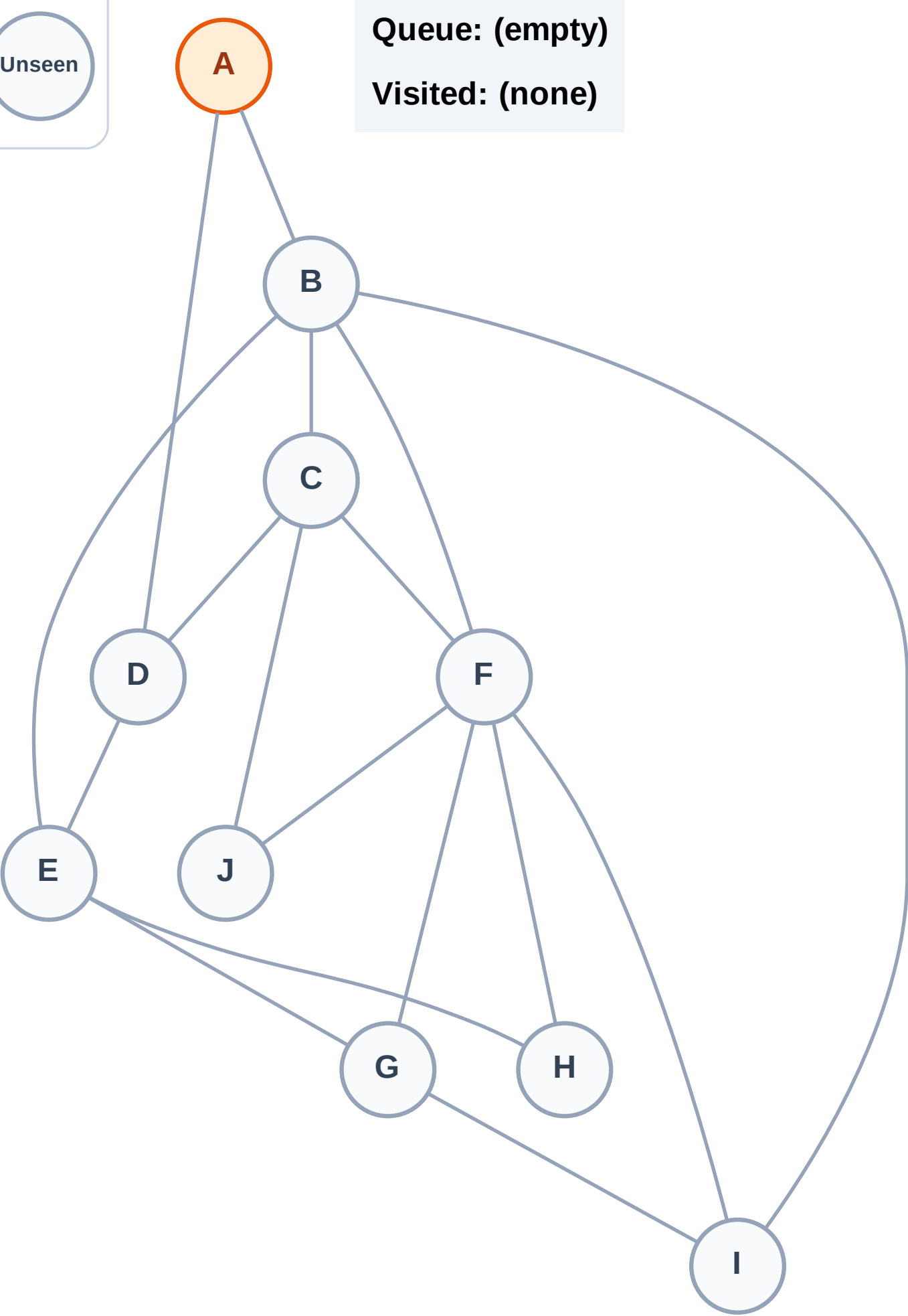
Complete

Processing

Discovered

Unseen

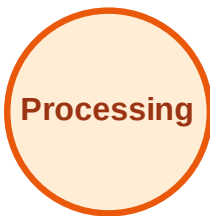
Queue: (empty)
Visited: (none)



BFS Traversal

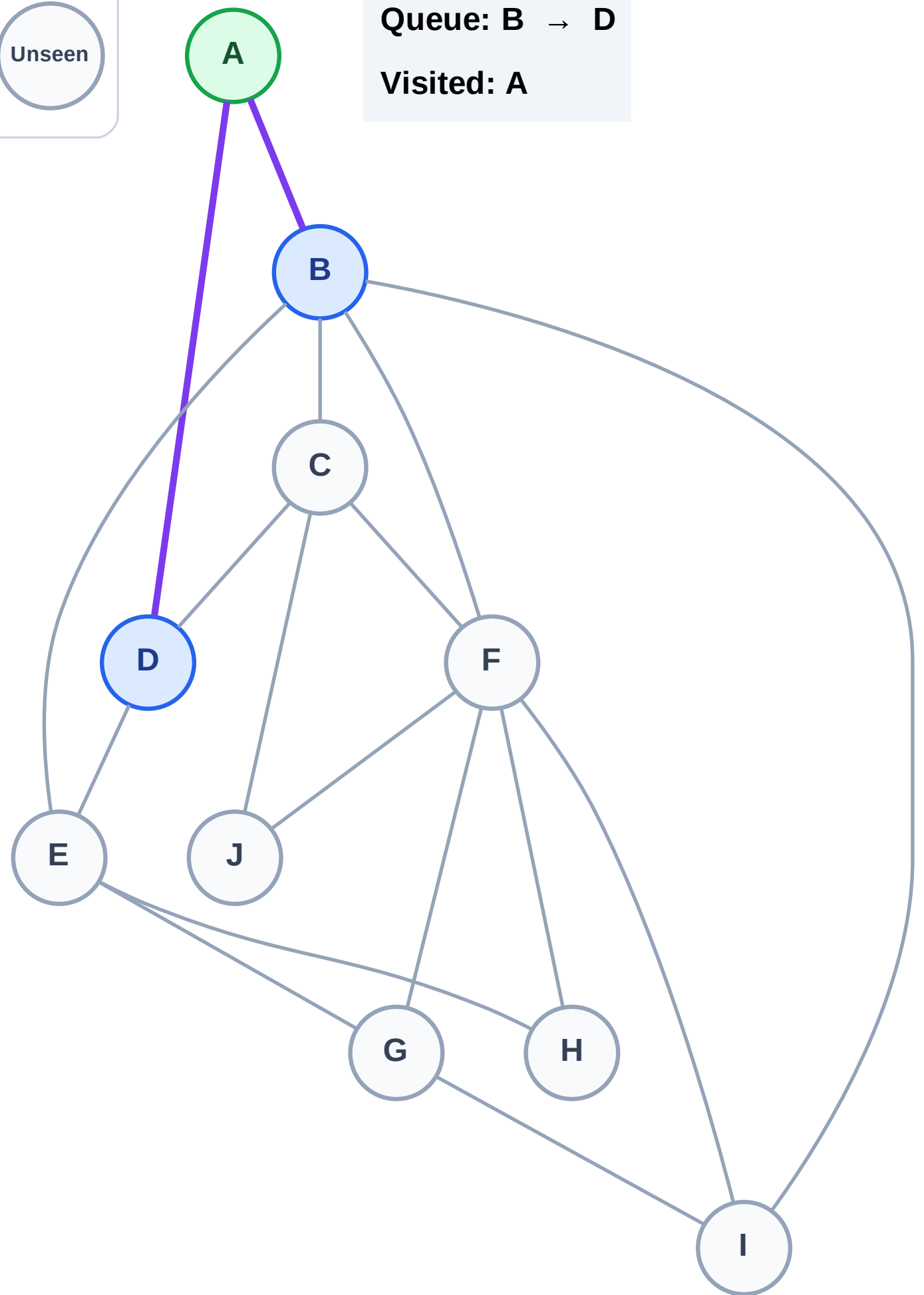
Step 3: Discover B, D from A

Legend



Queue: B → D

Visited: A



BFS Traversal

Step 4: Process node B

Legend

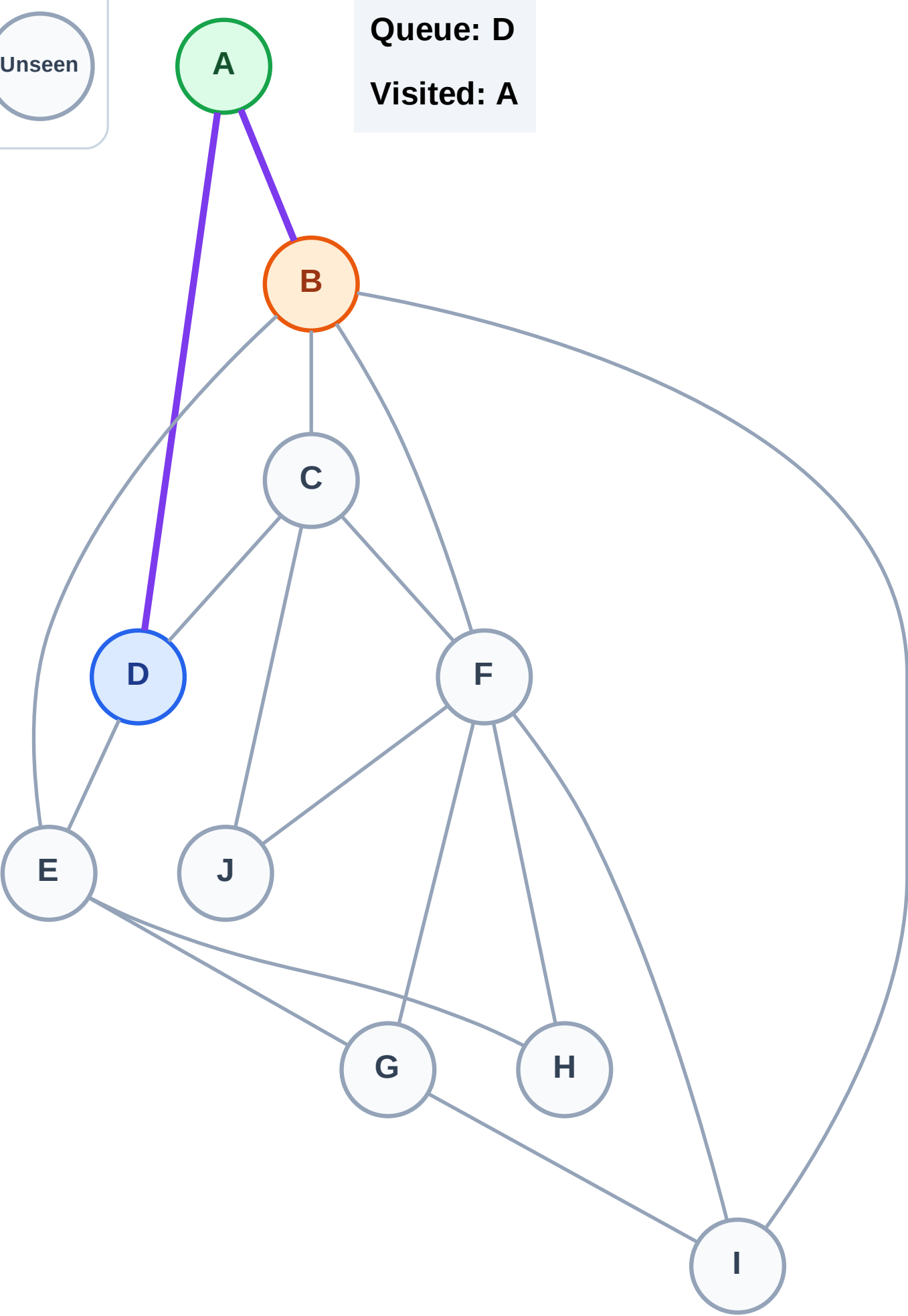
Complete

Processing

Discovered

Unseen

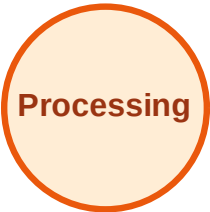
Queue: D
Visited: A



BFS Traversal

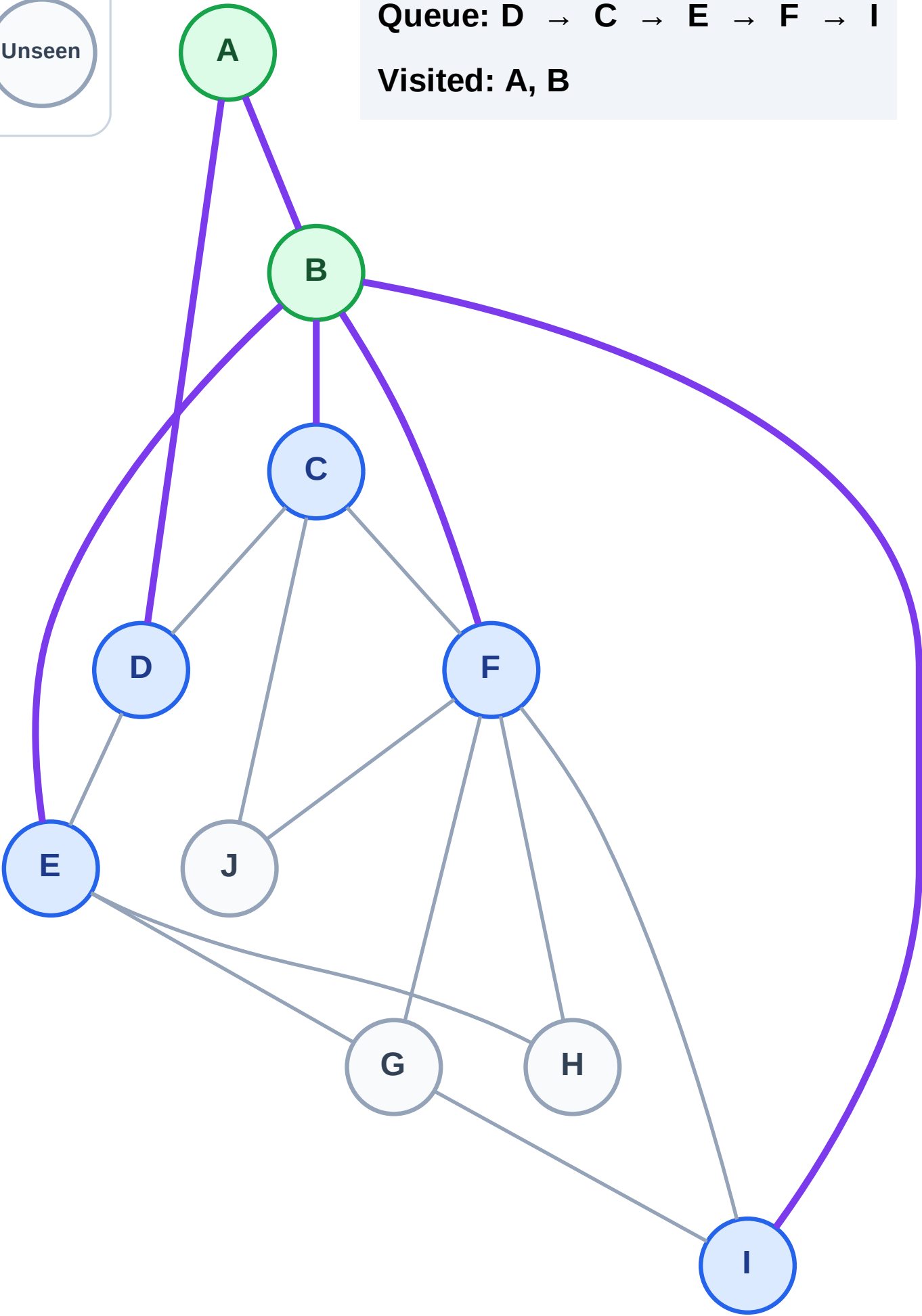
Step 5: Discover C, E, F, I from B

Legend



Queue: D → C → E → F → I

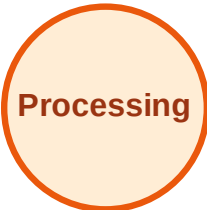
Visited: A, B



BFS Traversal

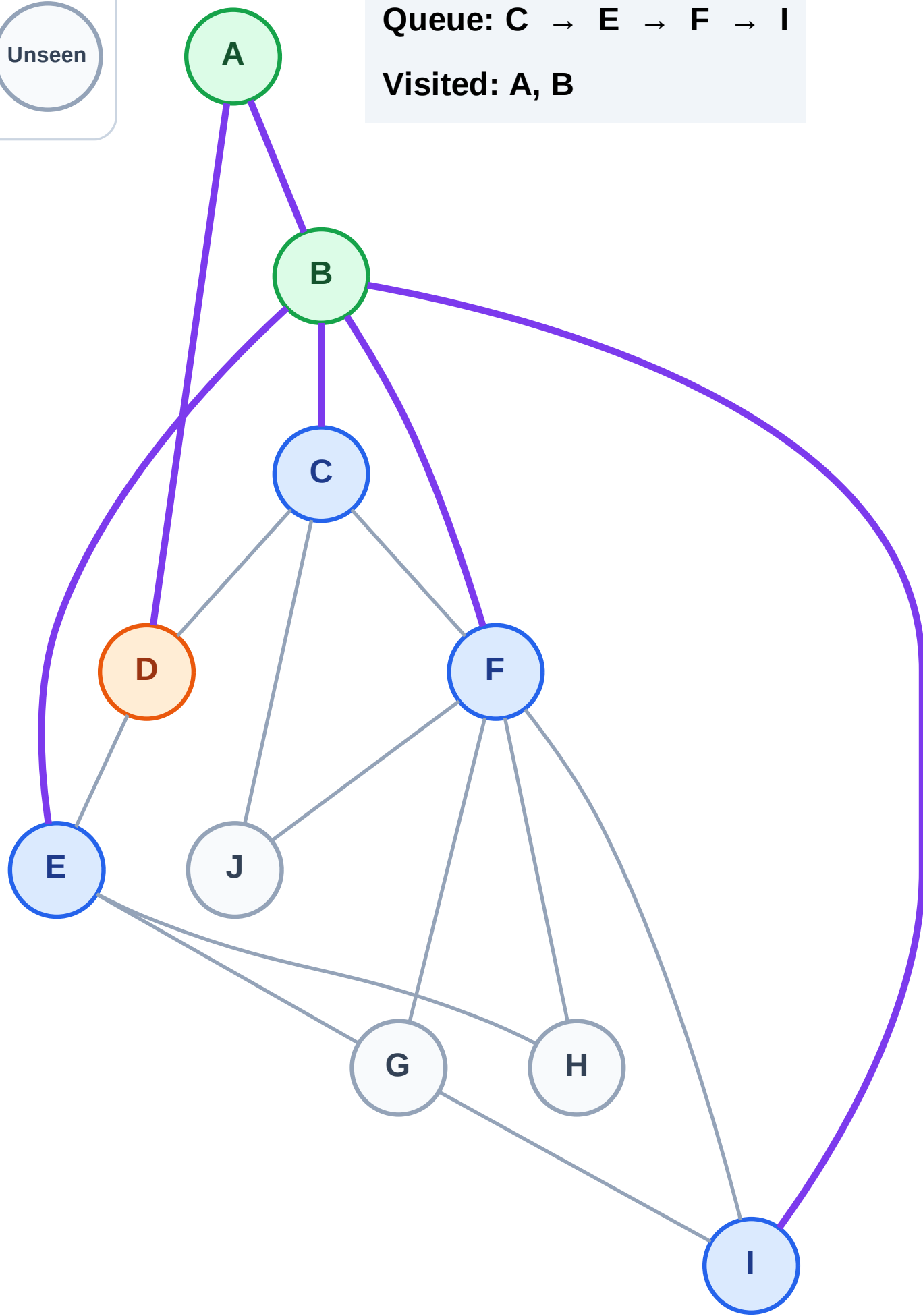
Step 6: Process node D

Legend



Queue: C → E → F → I

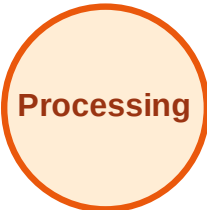
Visited: A, B



BFS Traversal

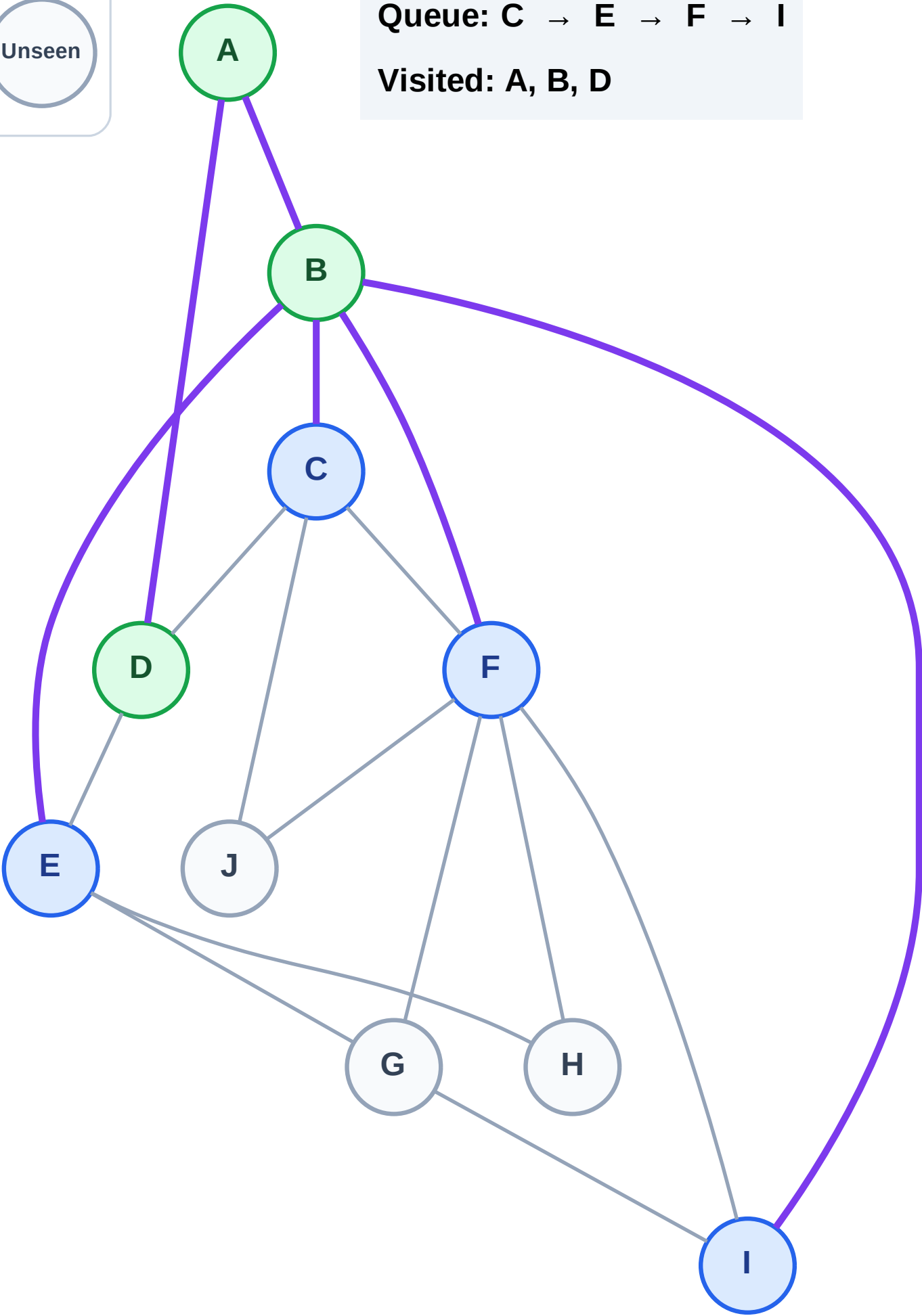
Step 7: No new neighbors from D

Legend



Queue: C → E → F → I

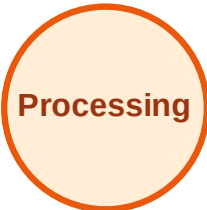
Visited: A, B, D



BFS Traversal

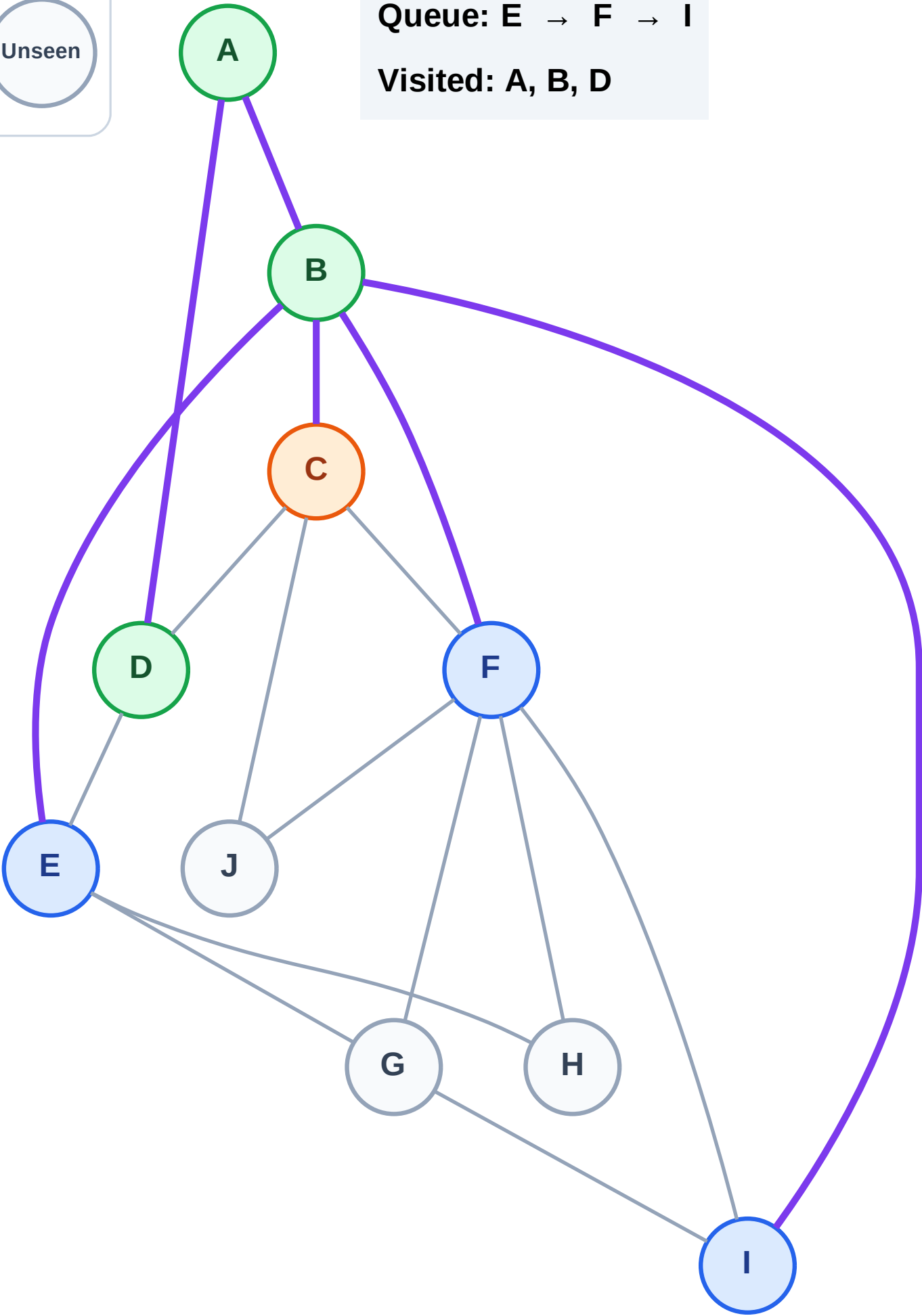
Step 8: Process node C

Legend



Queue: E → F → I

Visited: A, B, D



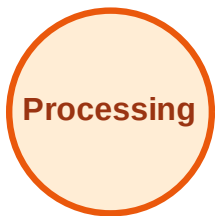
BFS Traversal

Step 9: Discover J from C

Legend



Complete



Processing



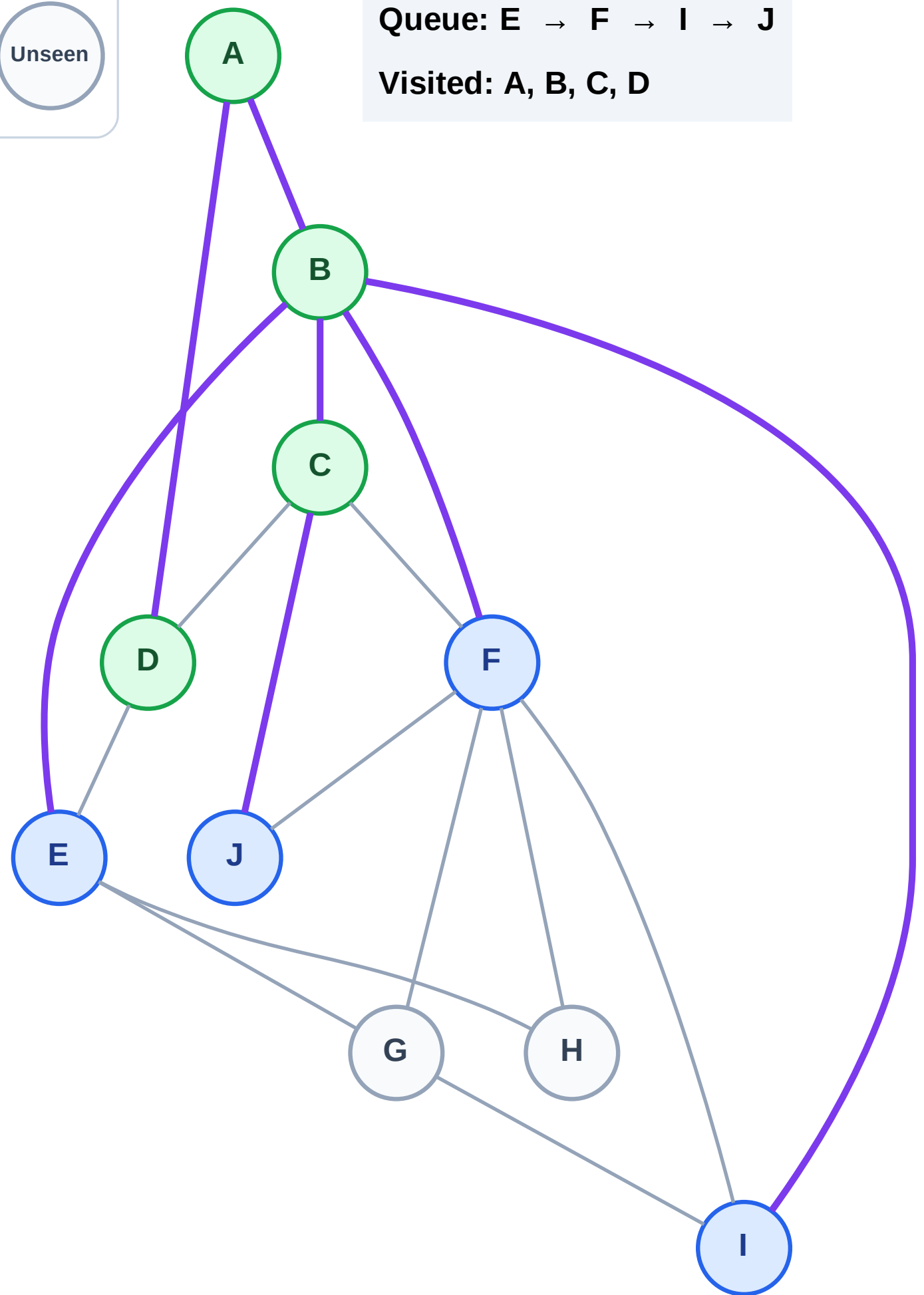
Discovered



Unseen

Queue: E → F → I → J

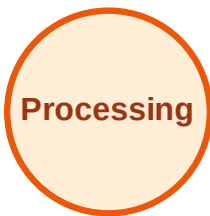
Visited: A, B, C, D



BFS Traversal

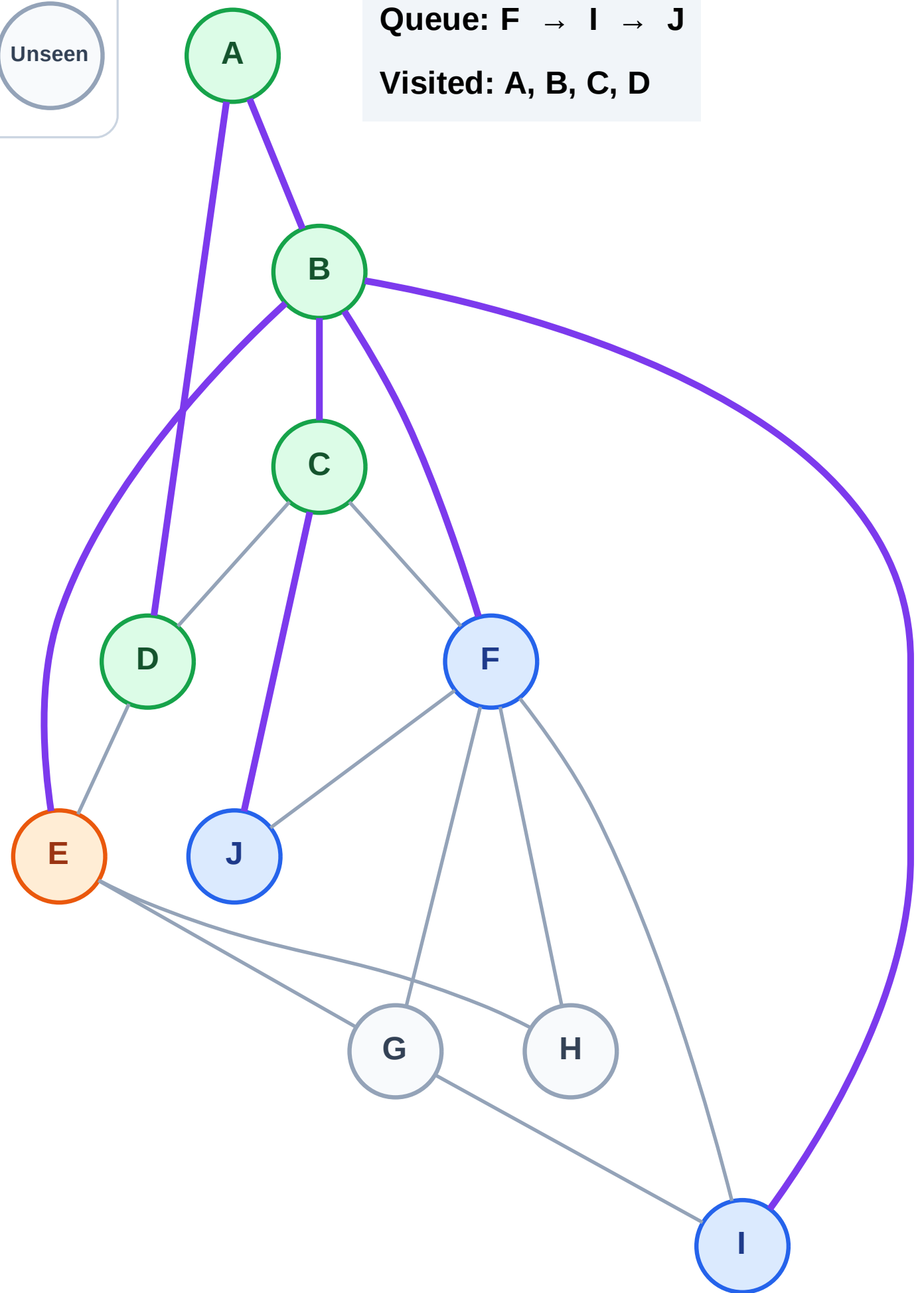
Step 10: Process node E

Legend



Queue: F → I → J

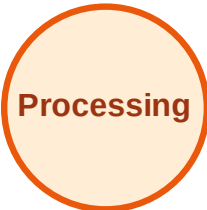
Visited: A, B, C, D



BFS Traversal

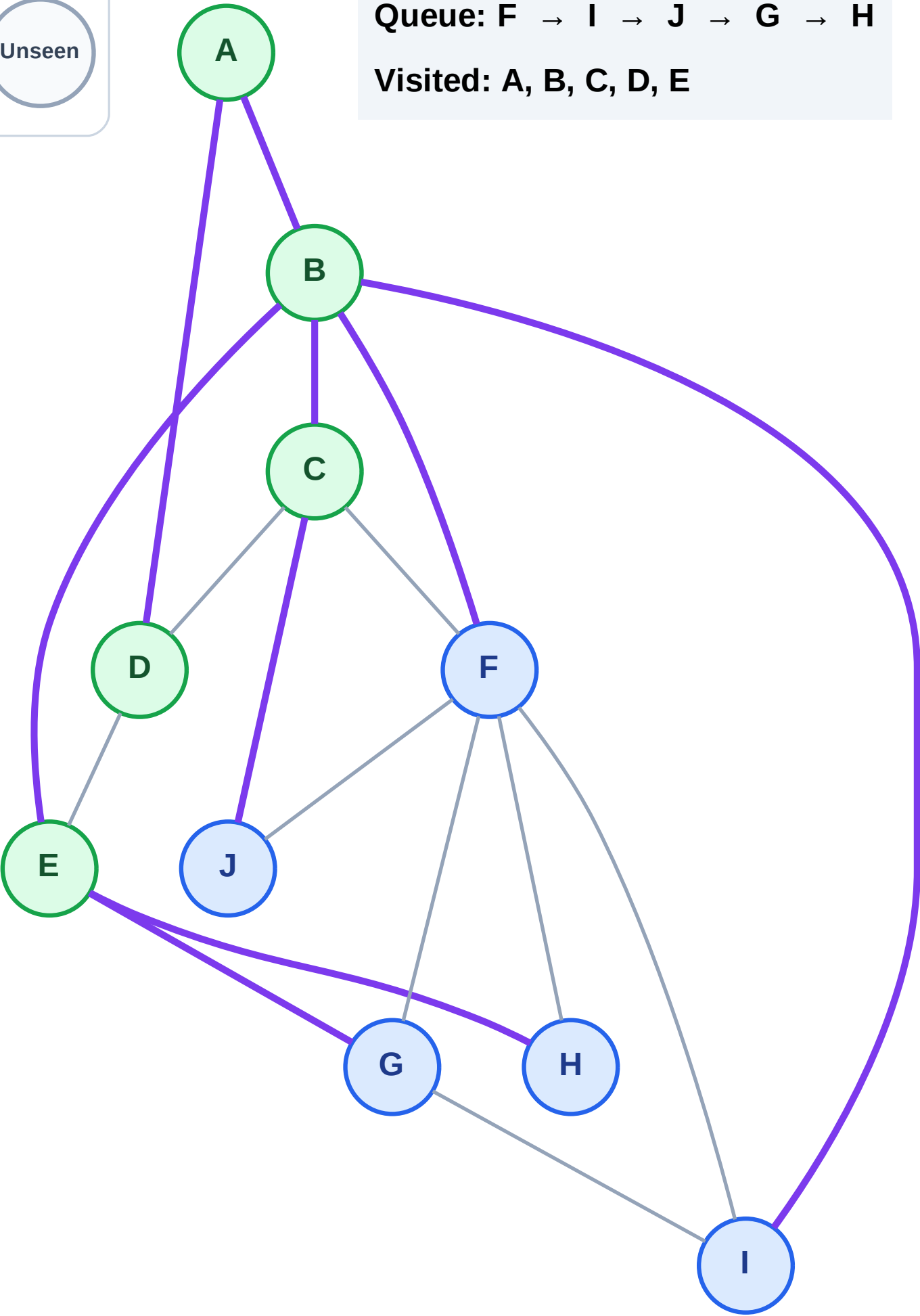
Step 11: Discover G, H from E

Legend



Queue: F → I → J → G → H

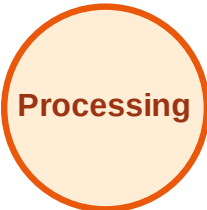
Visited: A, B, C, D, E



BFS Traversal

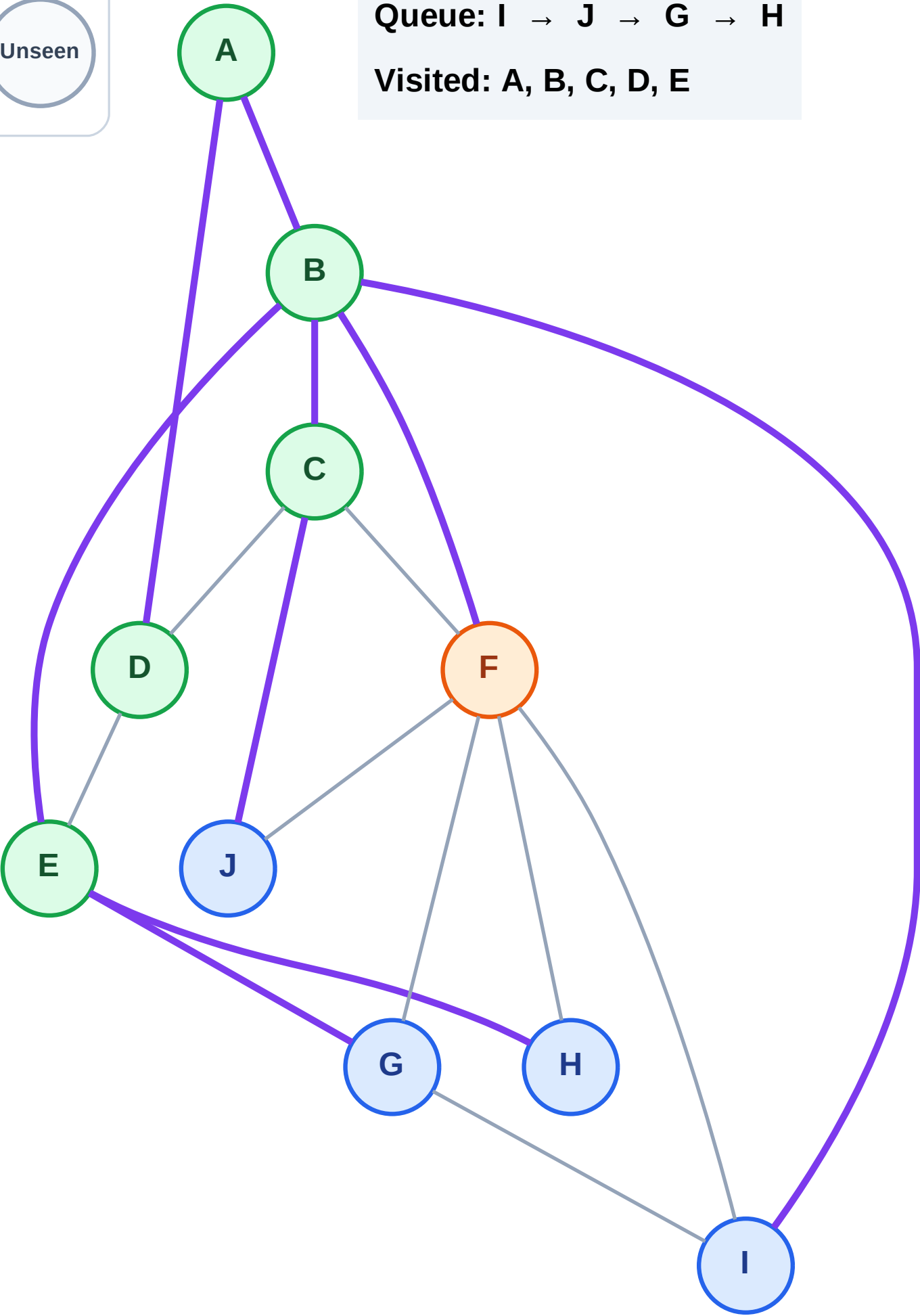
Step 12: Process node F

Legend



Queue: I → J → G → H

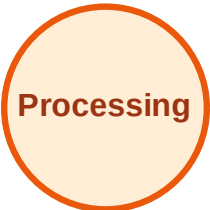
Visited: A, B, C, D, E



BFS Traversal

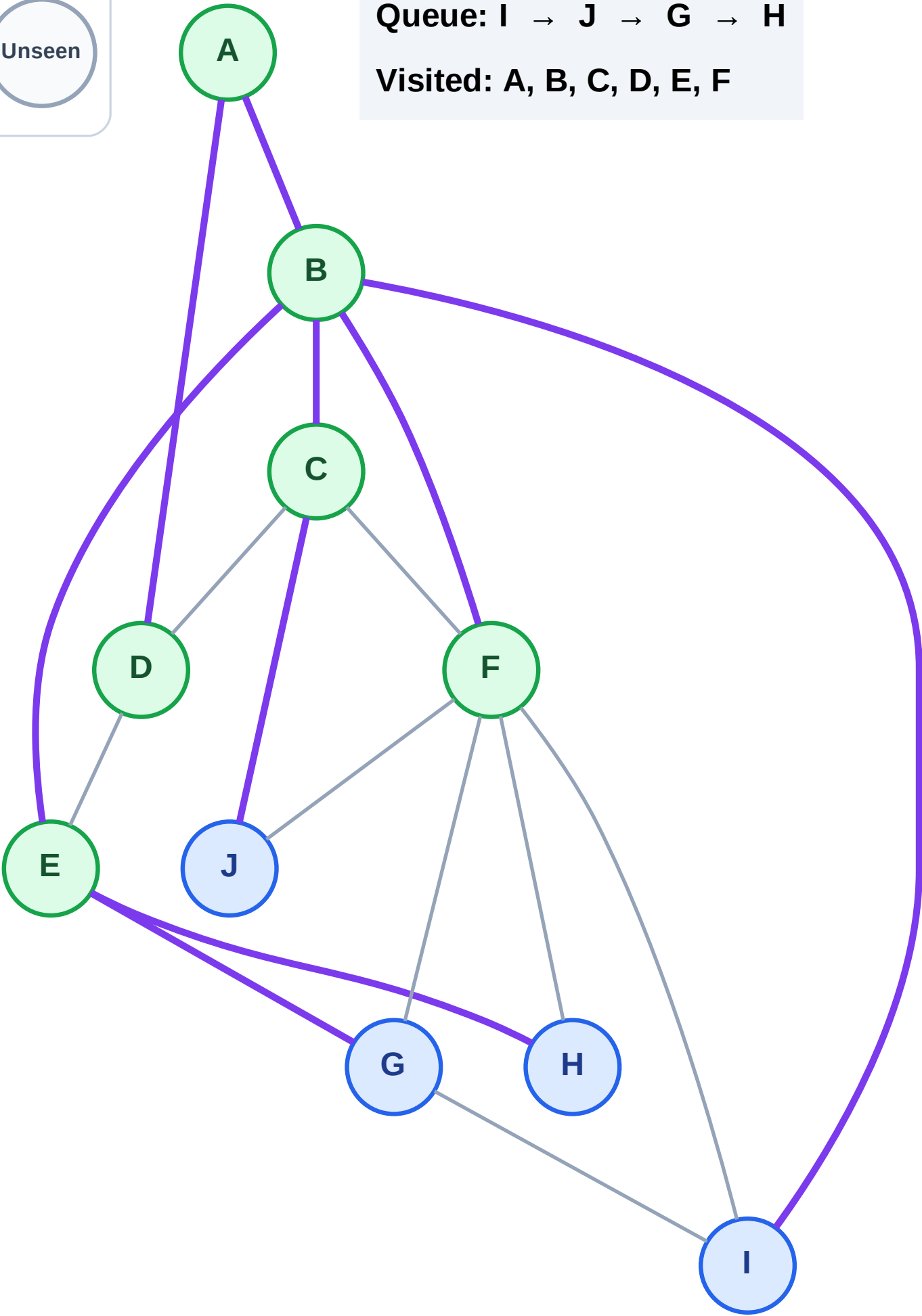
Step 13: No new neighbors from F

Legend



Queue: I → J → G → H

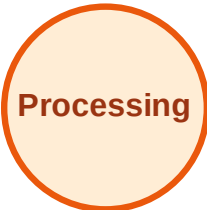
Visited: A, B, C, D, E, F



BFS Traversal

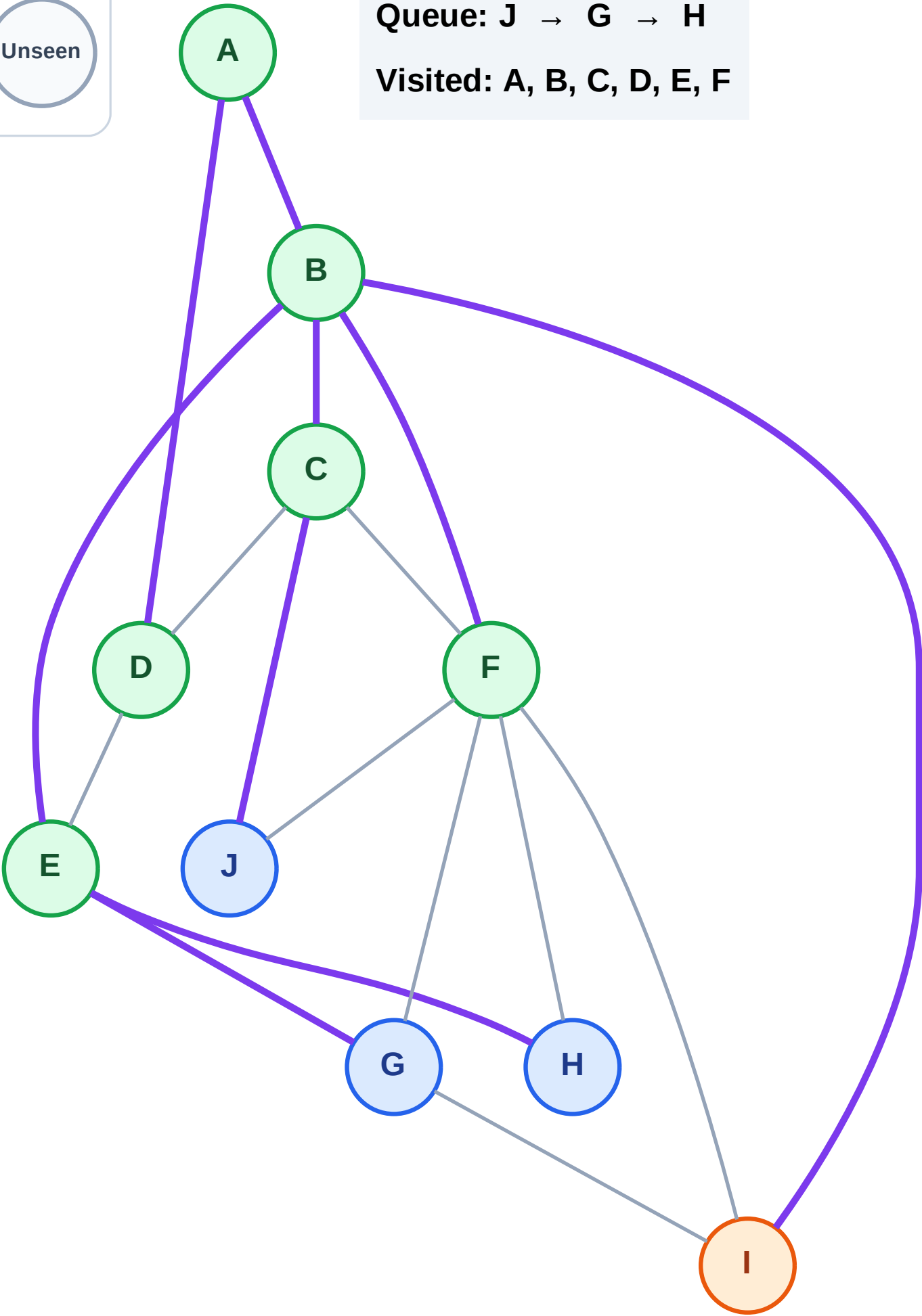
Step 14: Process node I

Legend



Queue: J → G → H

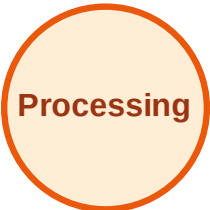
Visited: A, B, C, D, E, F



BFS Traversal

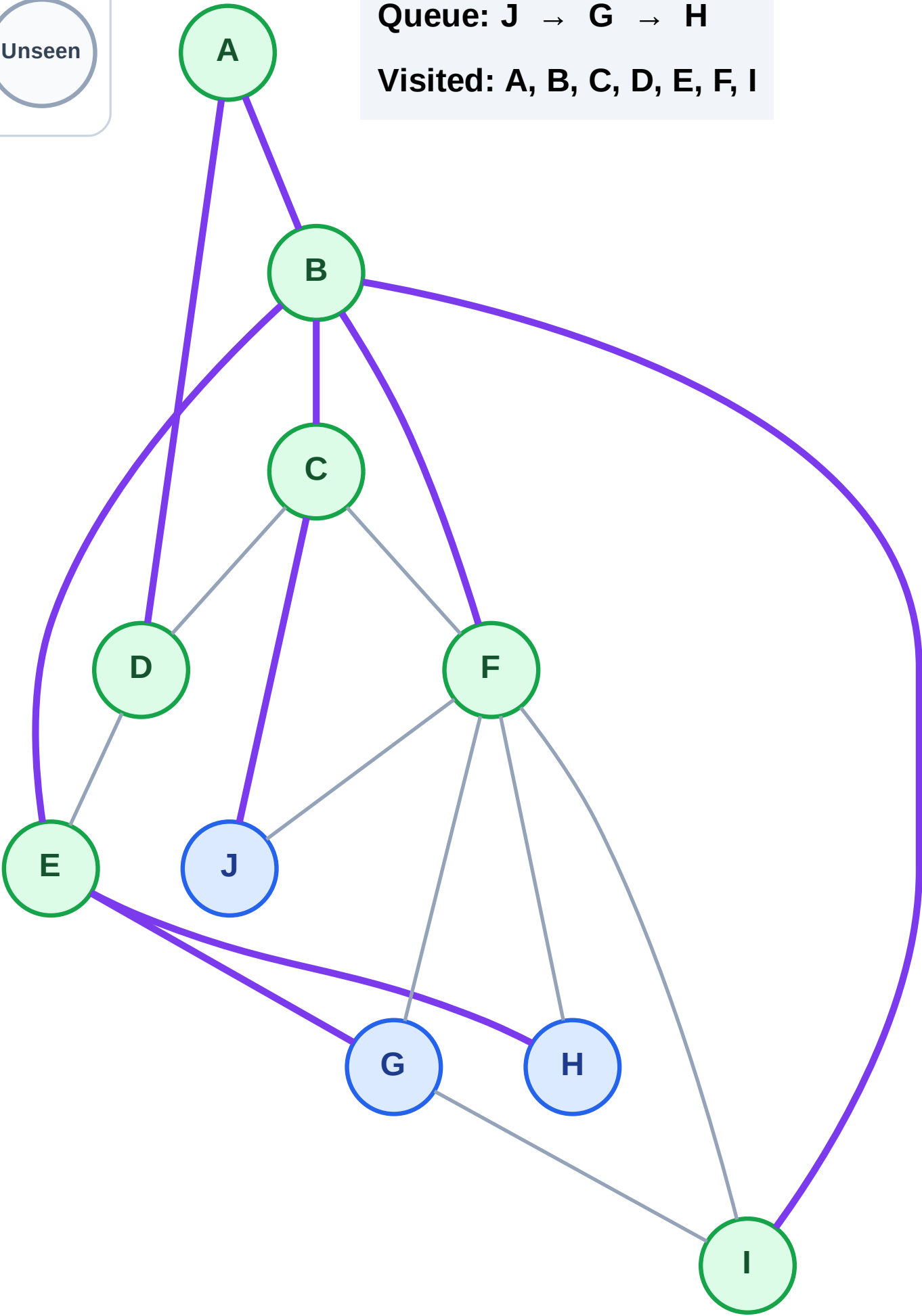
Step 15: No new neighbors from I

Legend



Queue: J → G → H

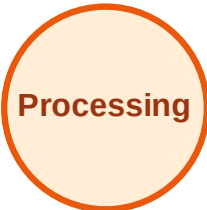
Visited: A, B, C, D, E, F, I



BFS Traversal

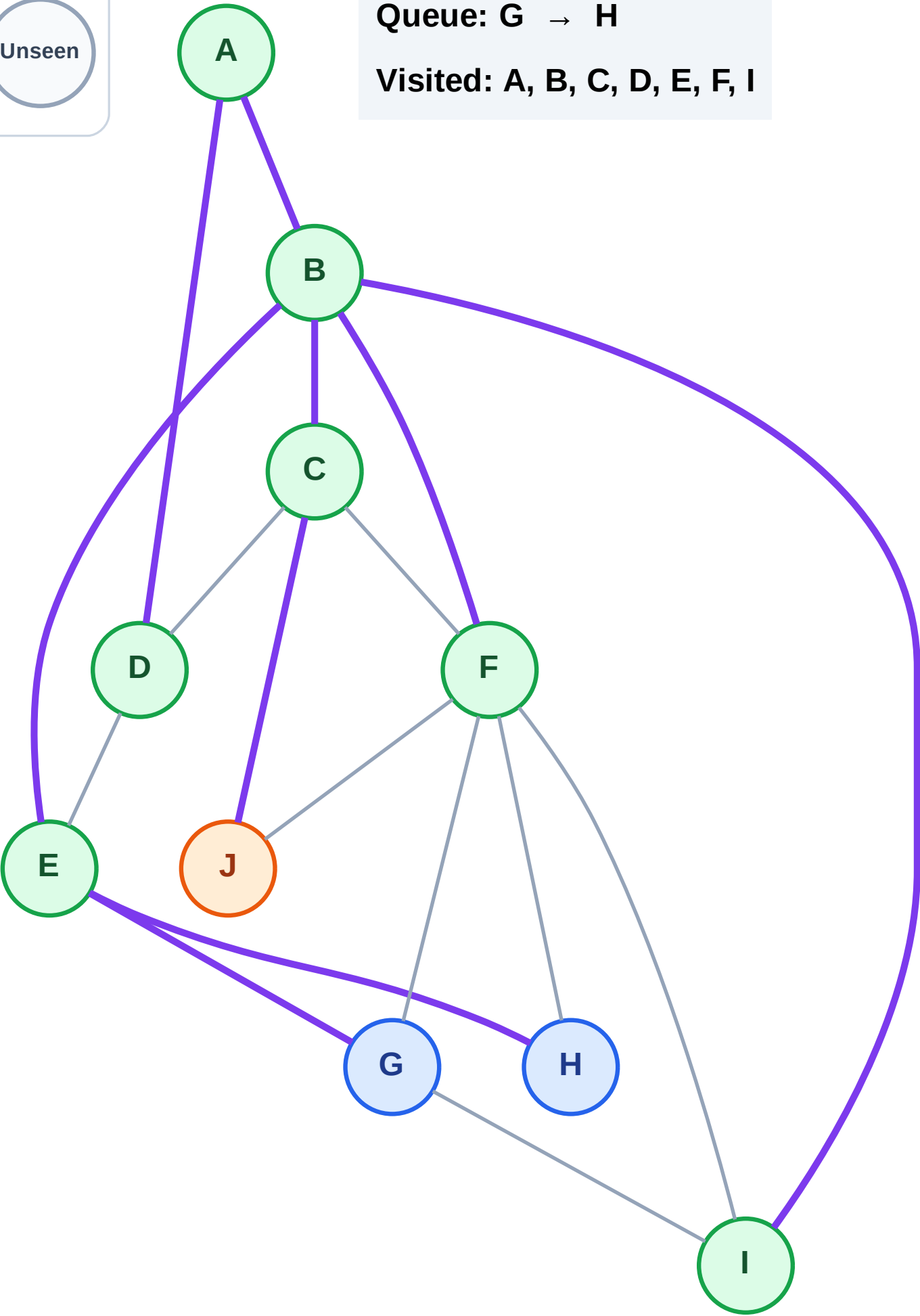
Step 16: Process node J

Legend



Queue: G → H

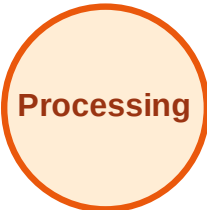
Visited: A, B, C, D, E, F, I



BFS Traversal

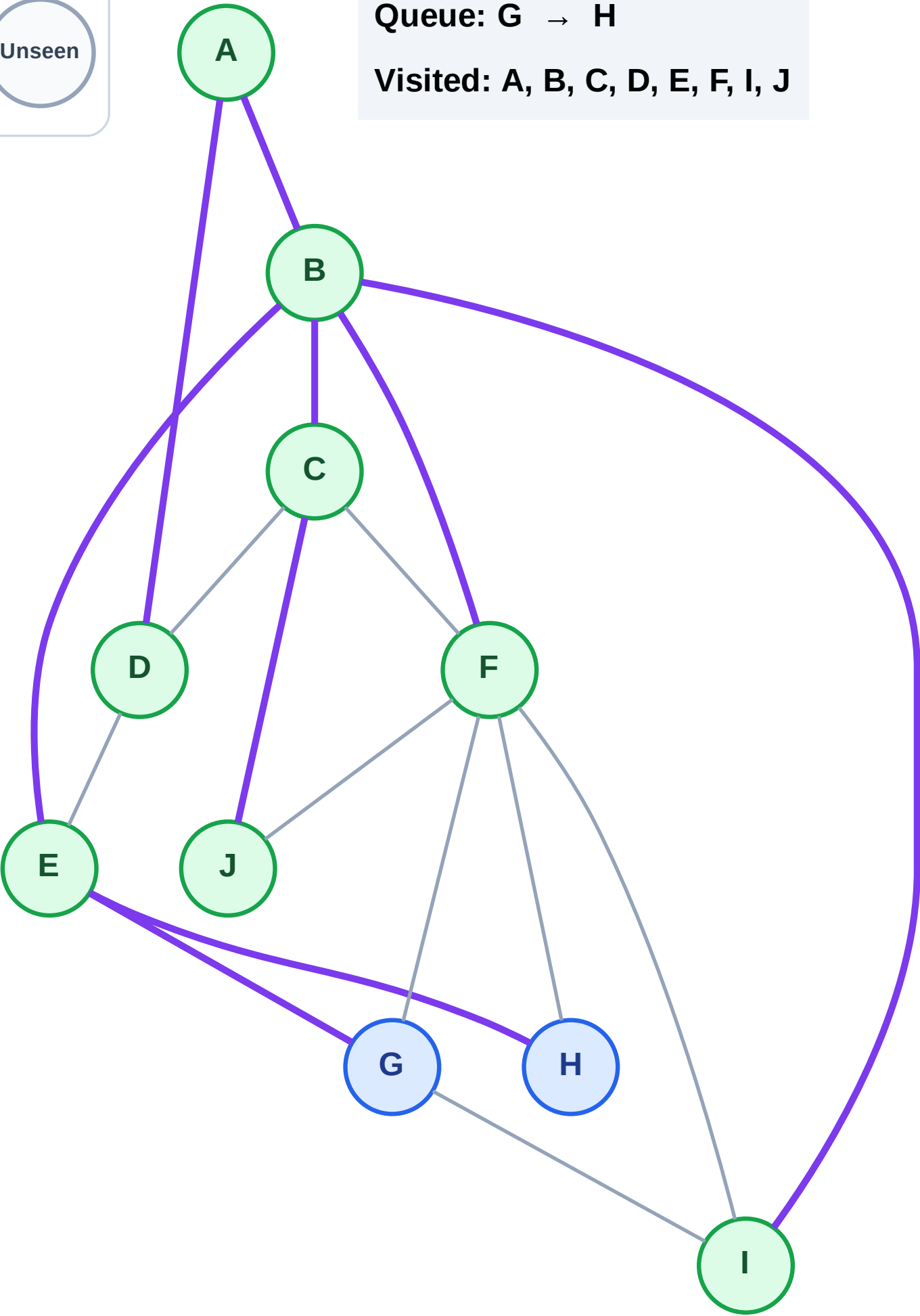
Step 17: No new neighbors from J

Legend



Queue: G → H

Visited: A, B, C, D, E, F, I, J



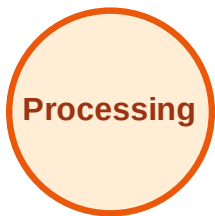
BFS Traversal

Step 18: Process node G

Legend



Complete



Processing



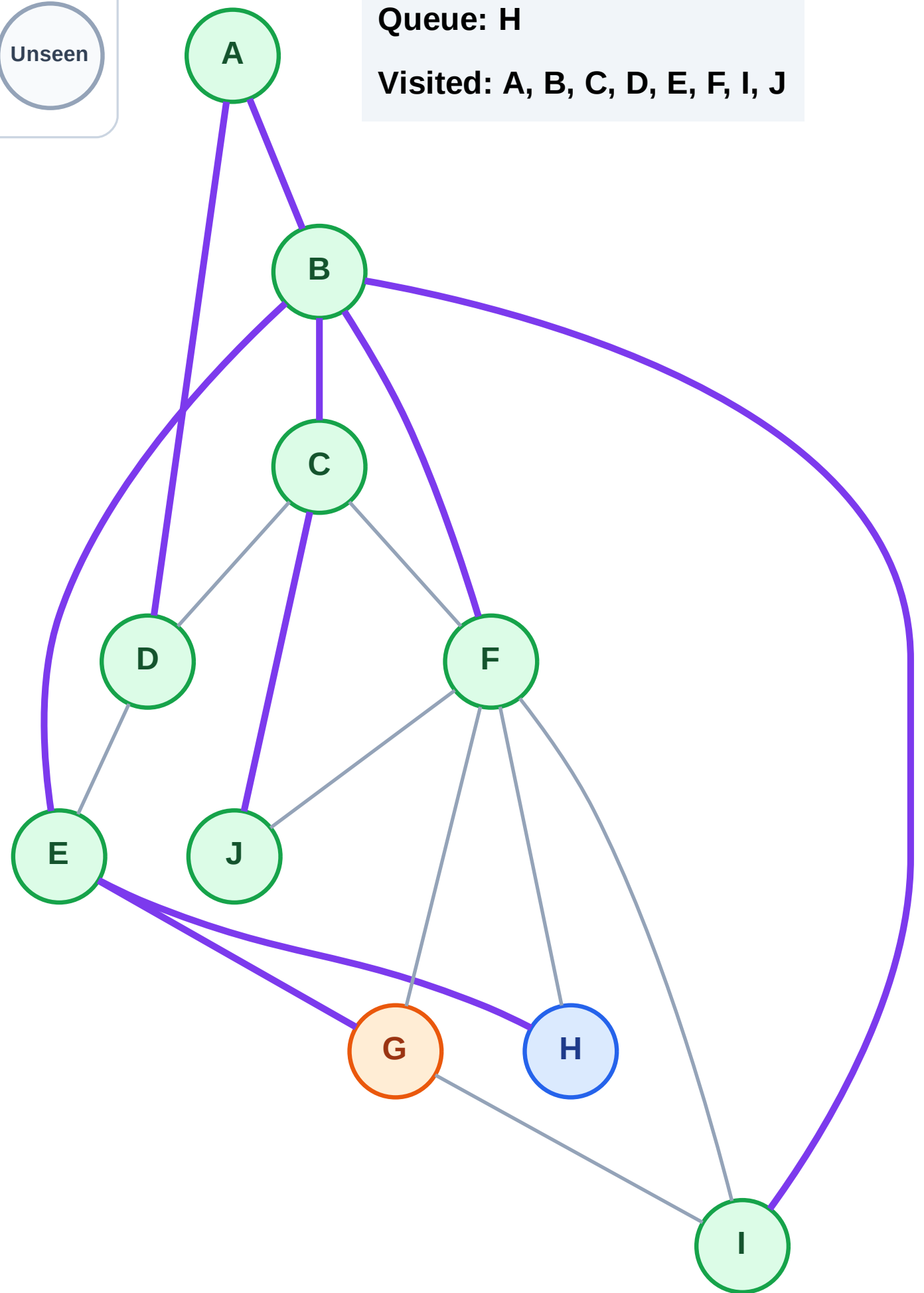
Discovered



Unseen

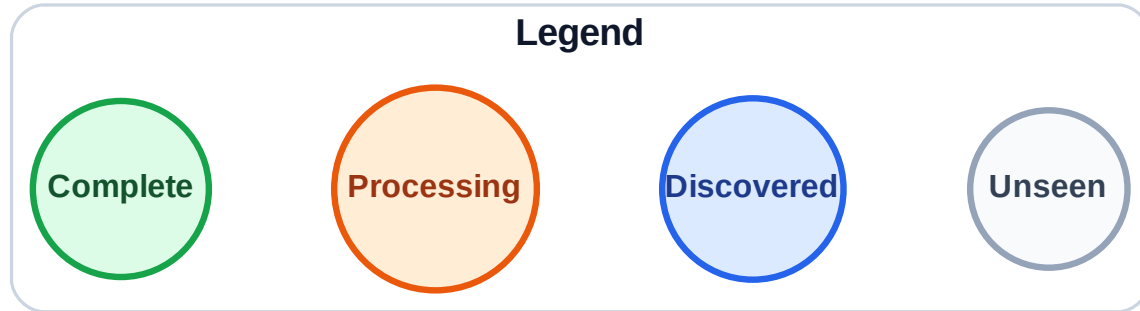
Queue: H

Visited: A, B, C, D, E, F, I, J



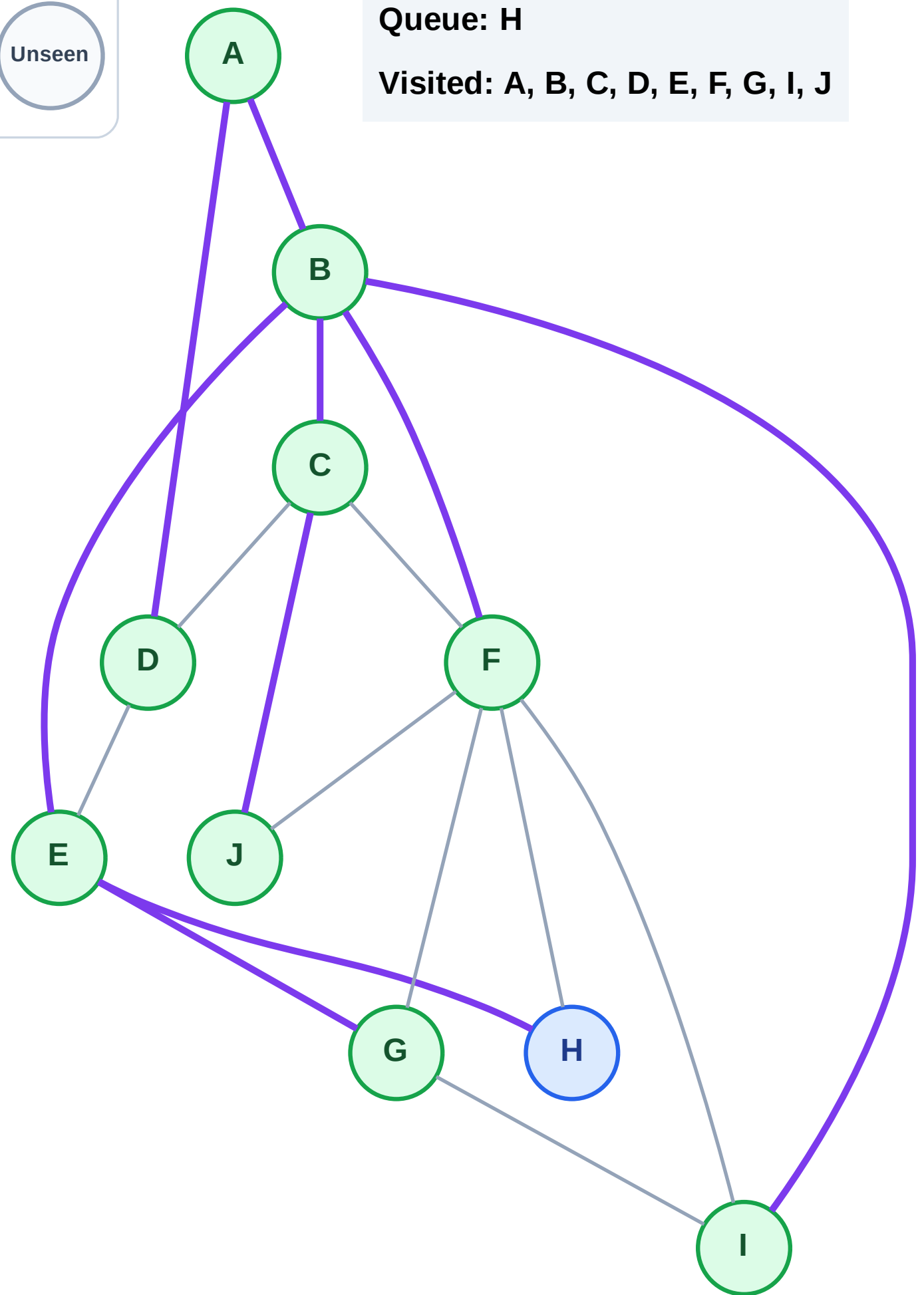
Step 19: No new neighbors from G

Step 19: No new neighbors from G



Queue: H

Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

Step 20: Process node H

Legend



Complete



Processing



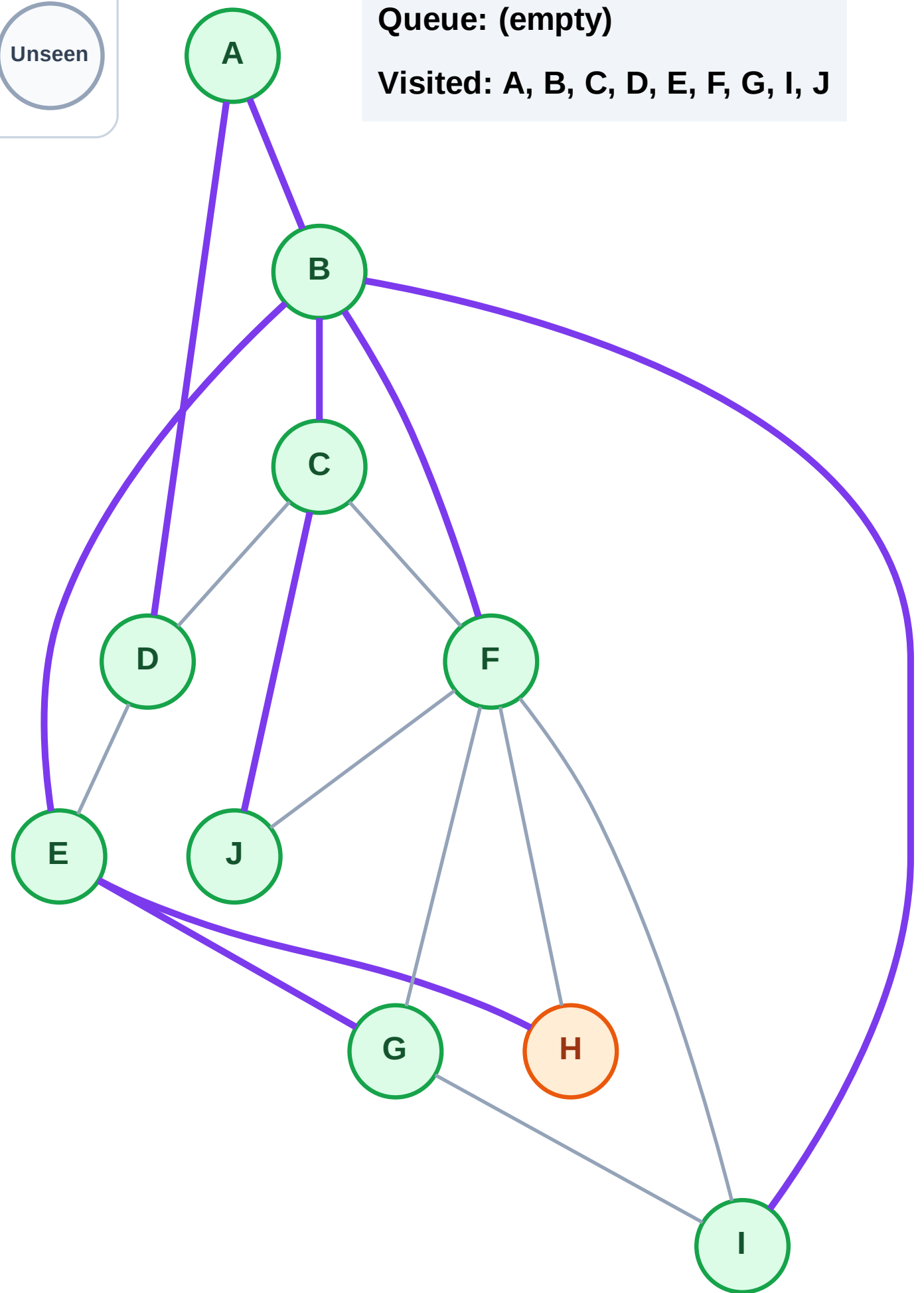
Discovered



Unseen

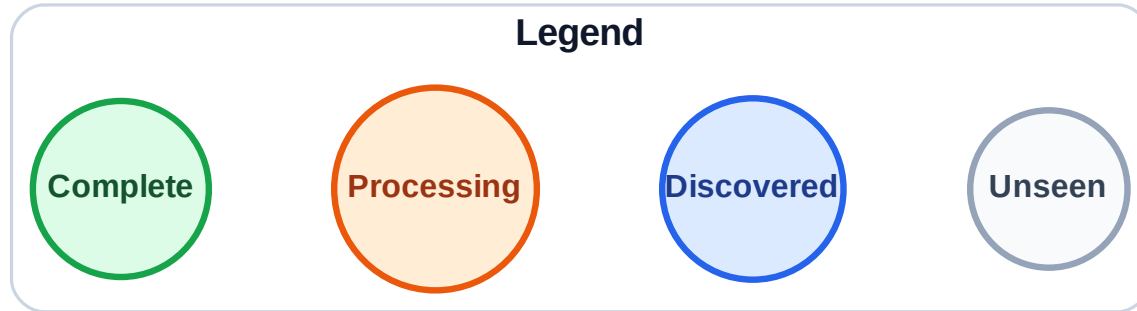
Queue: (empty)

Visited: A, B, C, D, E, F, G, I, J



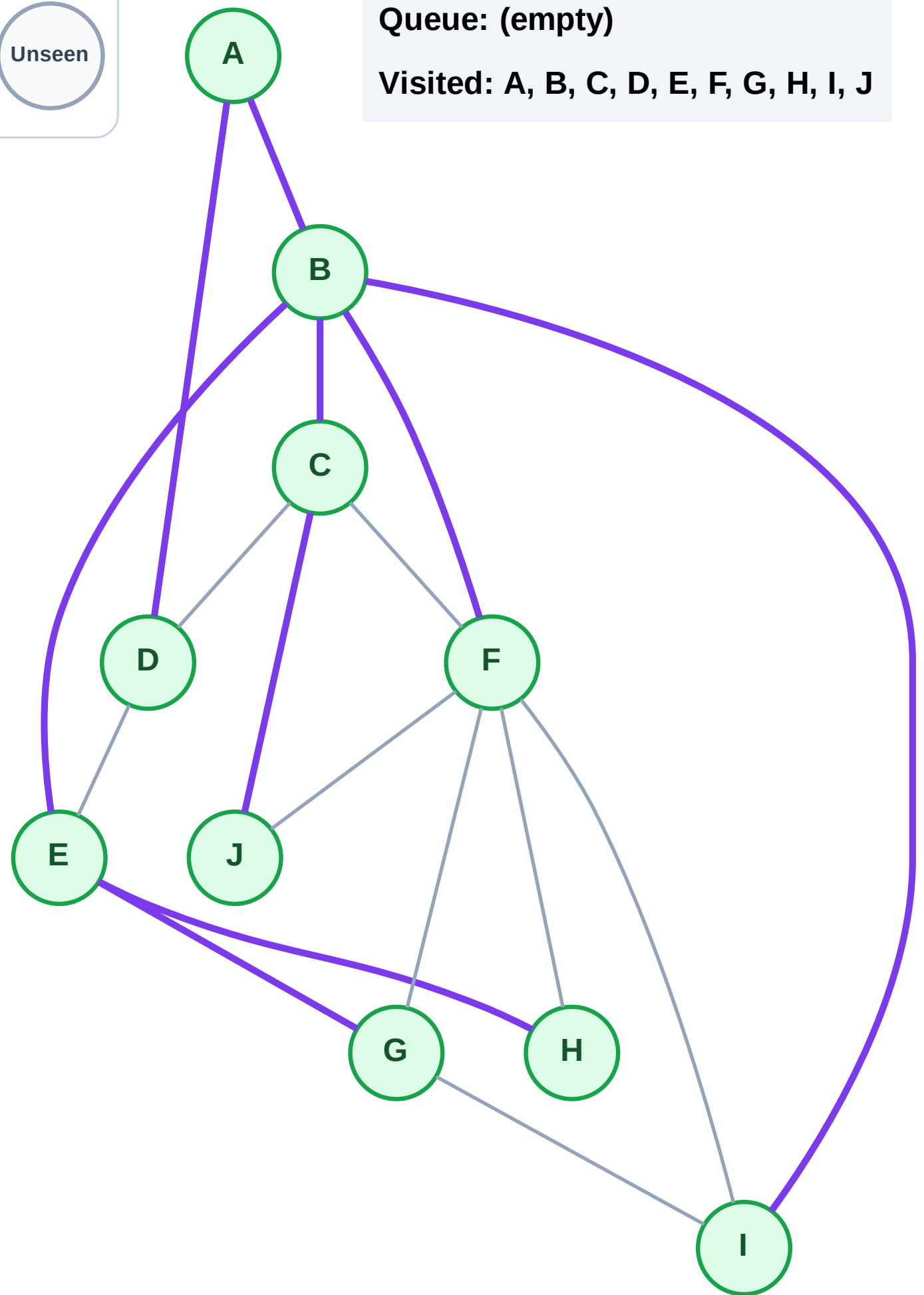
Step 21: No new neighbors from H

Step 21: No new neighbors from H



Queue: (empty)

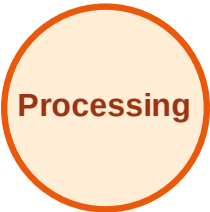
Visited: A, B, C, D, E, F, G, H, I, J



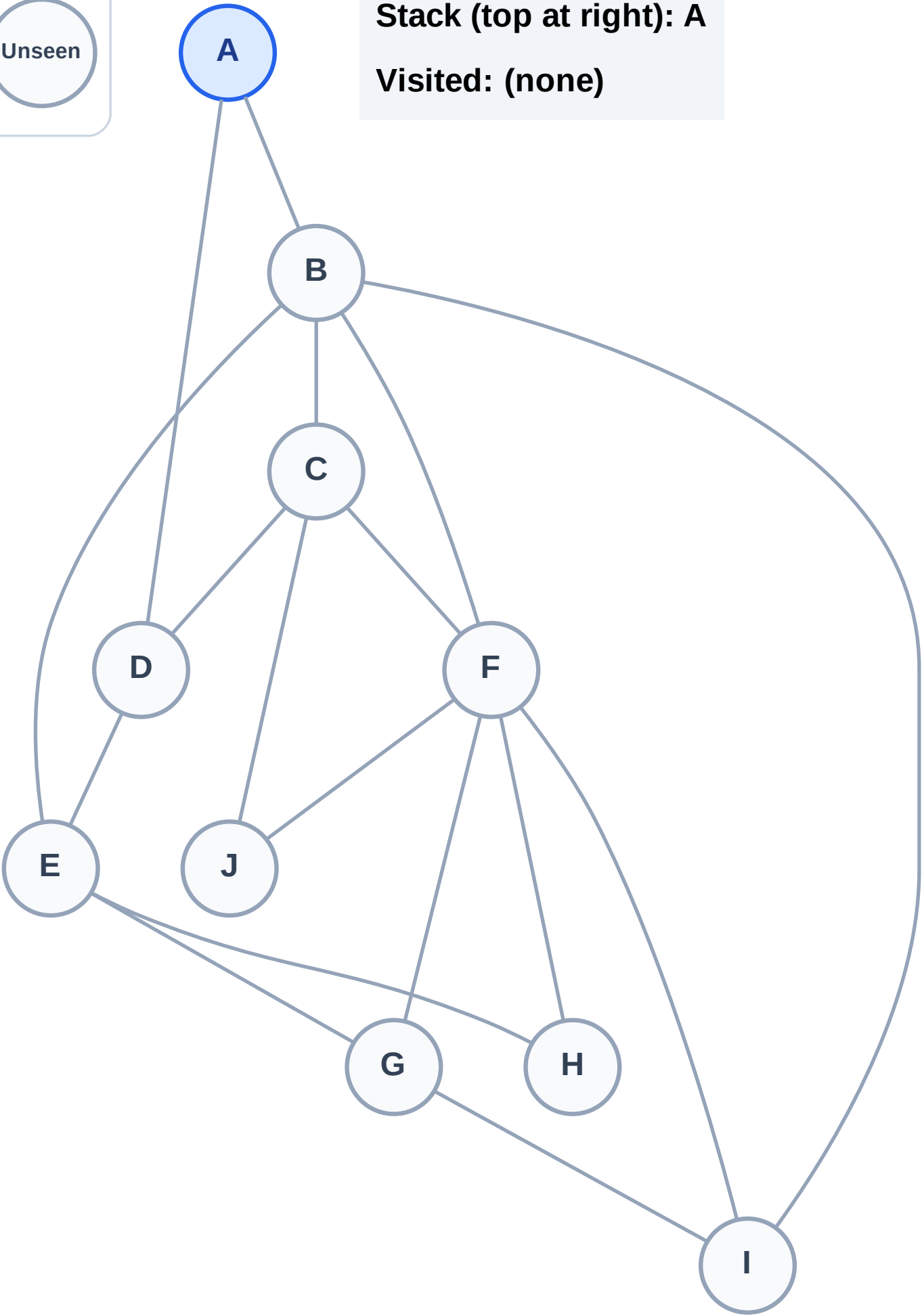
DFS Traversal

Step 1: Start at A

Legend



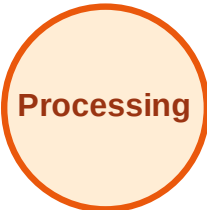
Stack (top at right): A
Visited: (none)



DFS Traversal

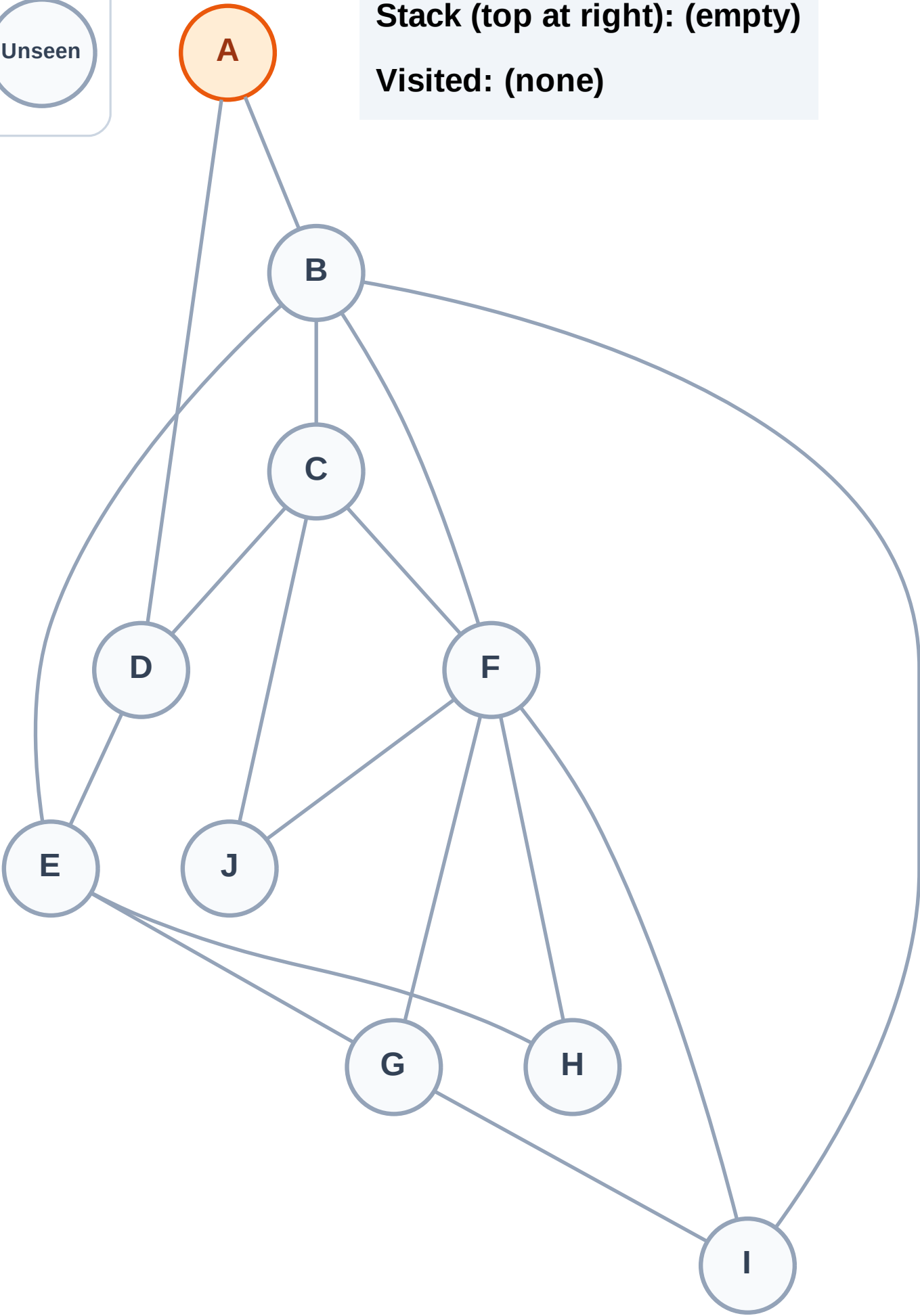
Step 2: Process node A

Legend



Stack (top at right): (empty)

Visited: (none)



DFS Traversal

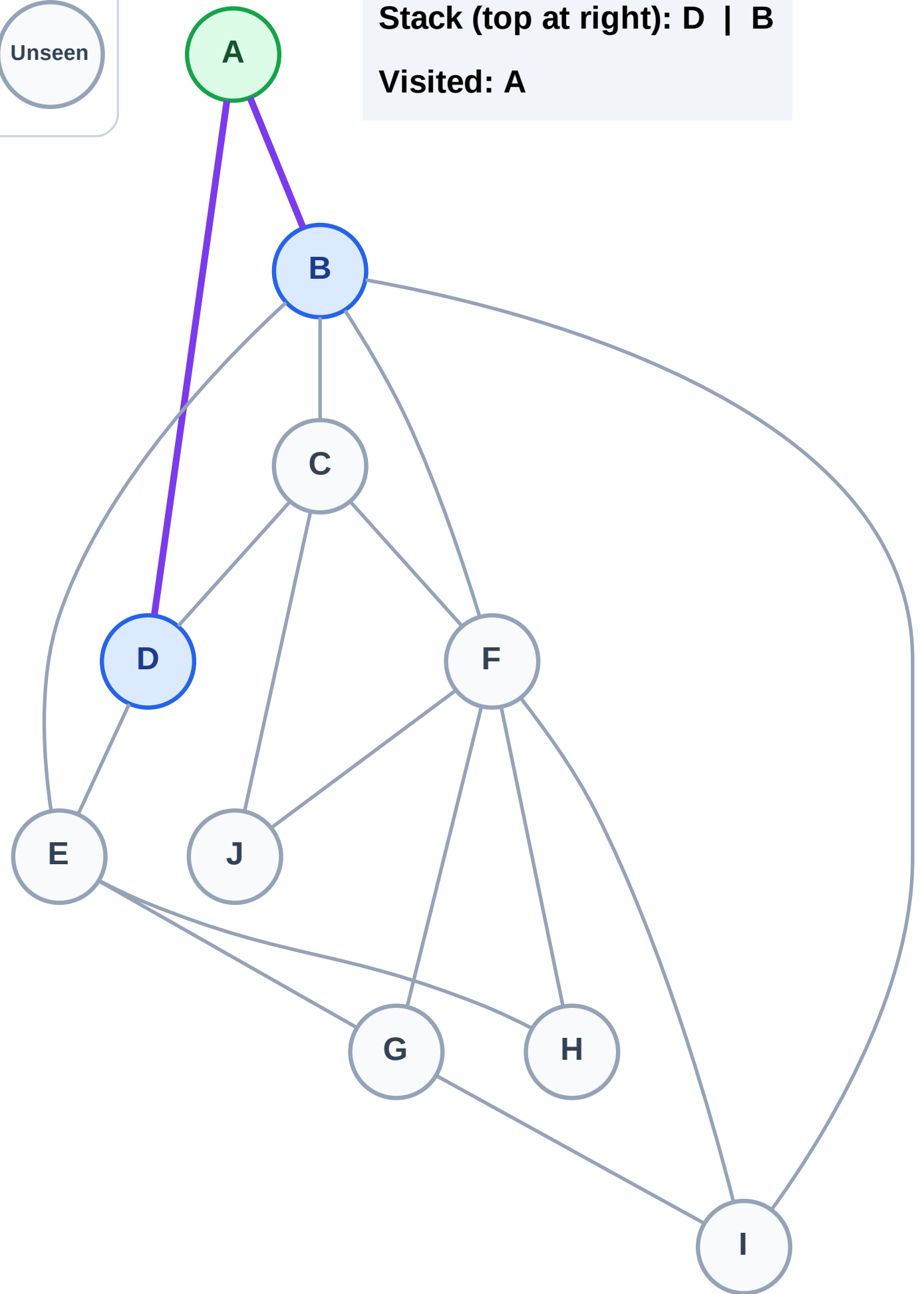
Step 3: Discover D, B from A

Legend



Stack (top at right): D | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

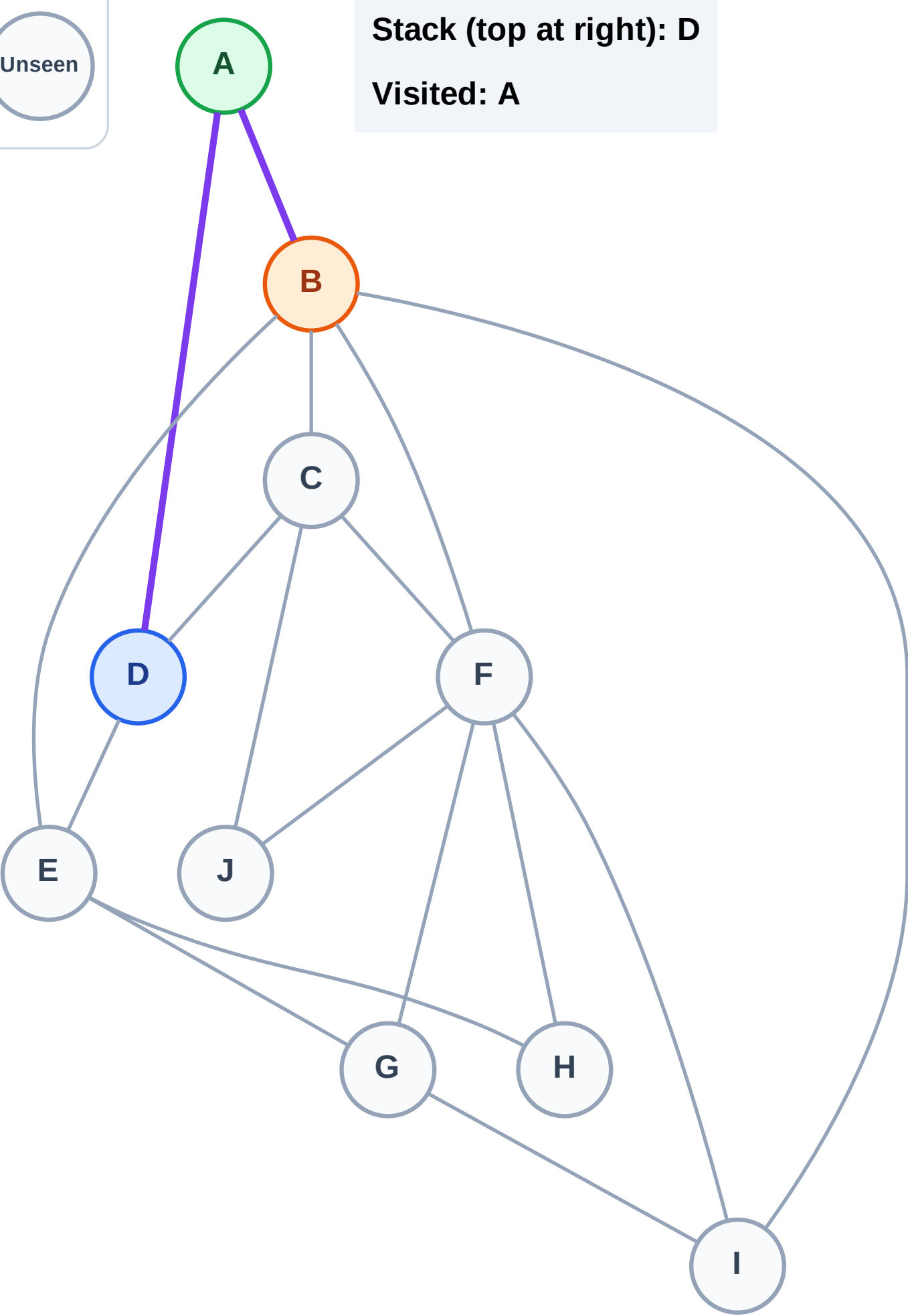
Complete

Processing

Discovered

Unseen

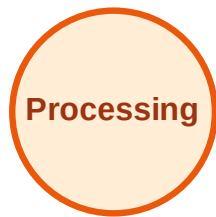
Stack (top at right): D
Visited: A



DFS Traversal

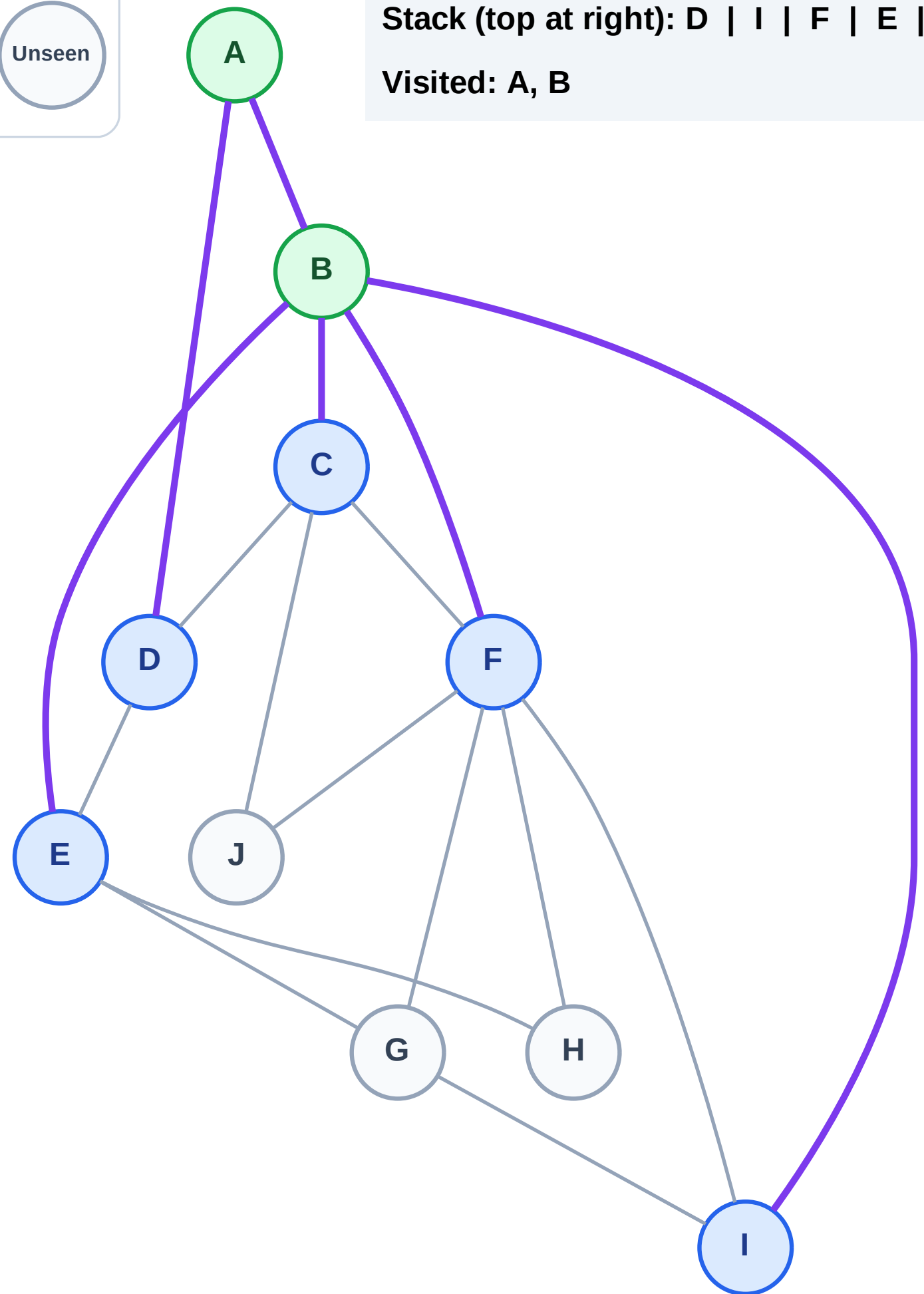
Step 5: Discover I, F, E, C from B

Legend



Stack (top at right): D | I | F | E | C

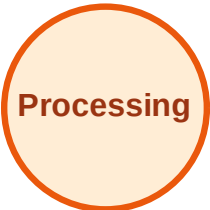
Visited: A, B



DFS Traversal

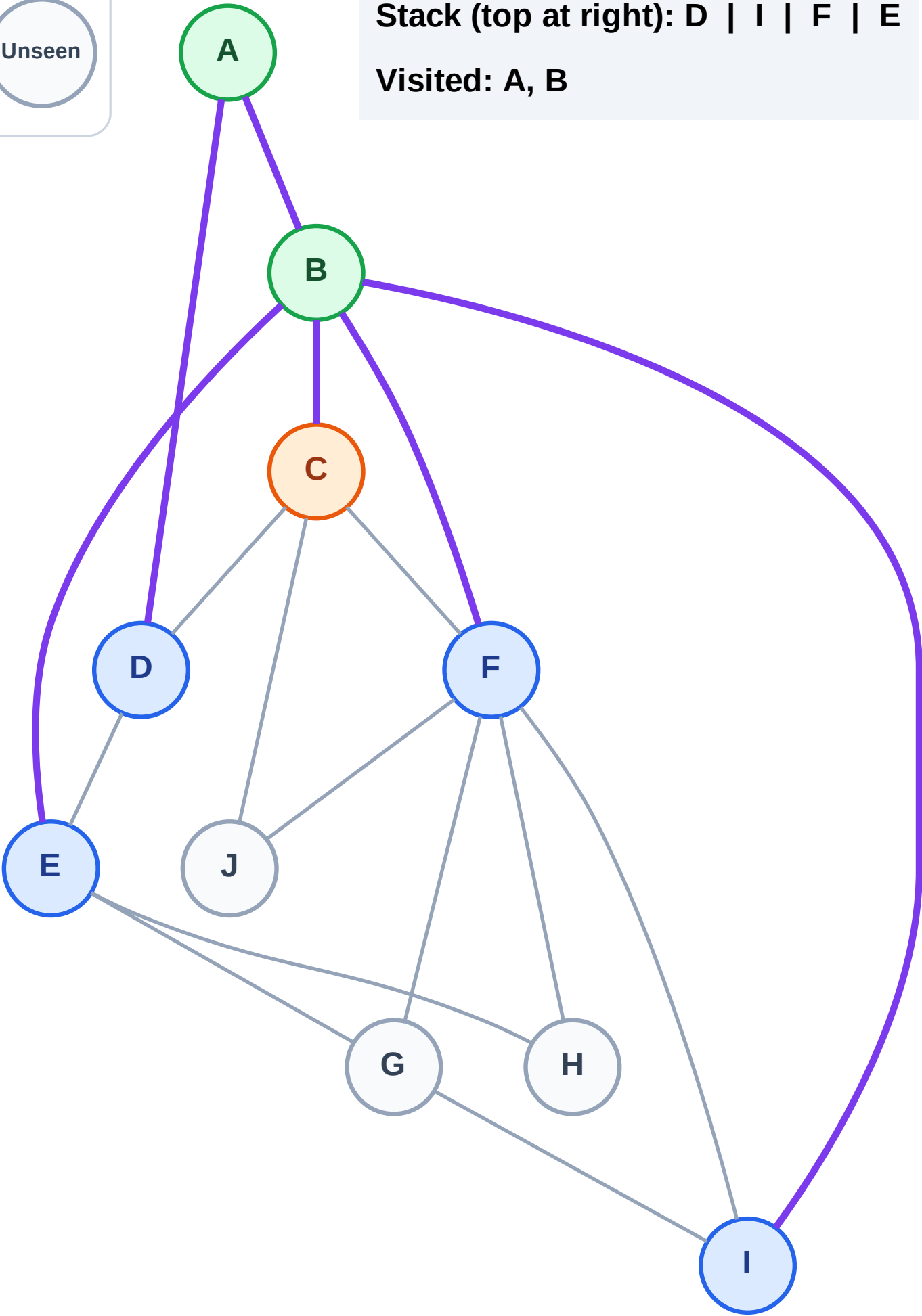
Step 6: Process node C

Legend



Stack (top at right): D | I | F | E

Visited: A, B



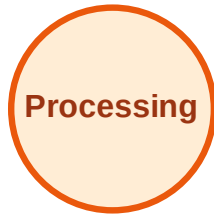
DFS Traversal

Step 7: Discover J from C

Legend



Complete



Processing



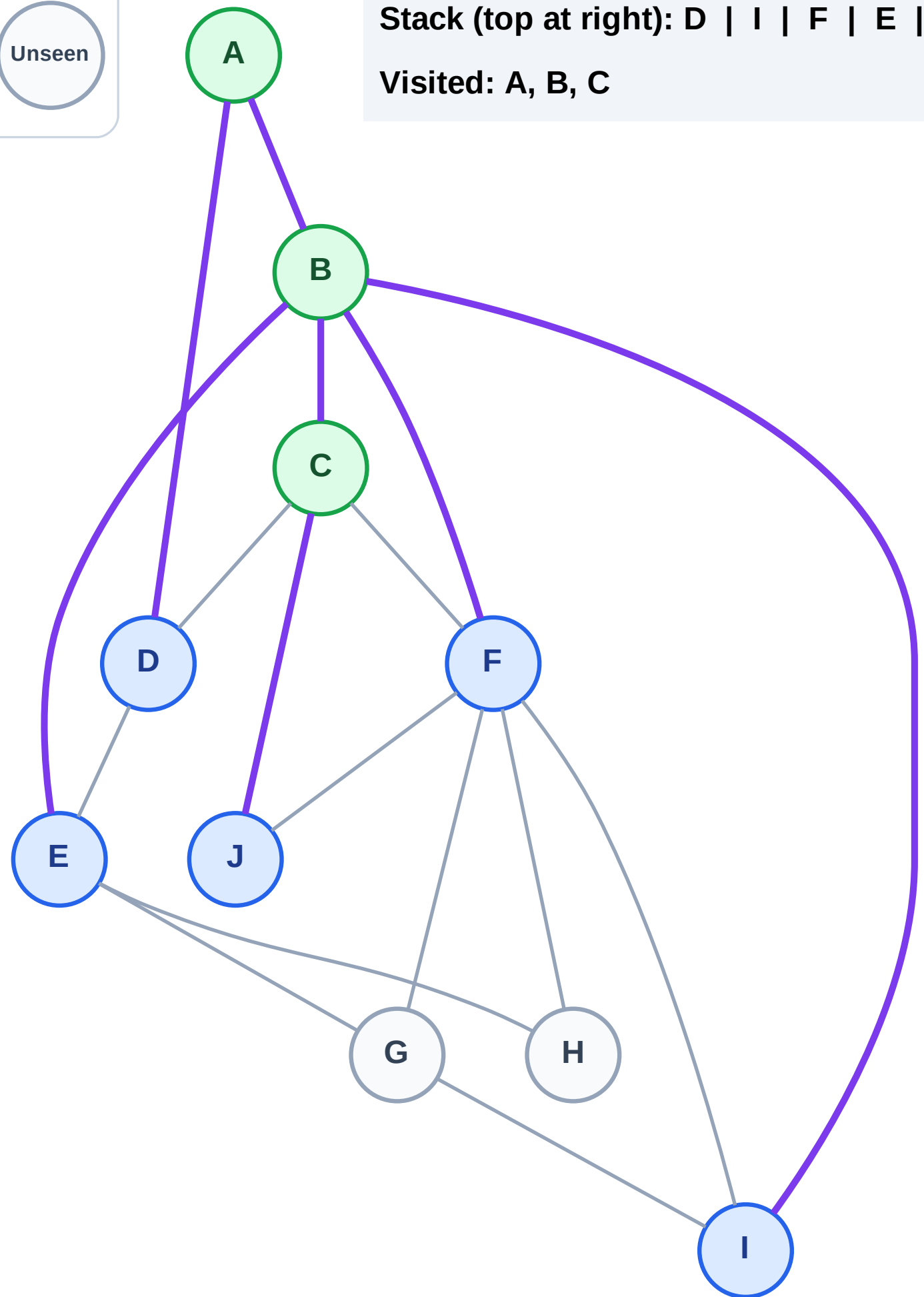
Discovered



Unseen

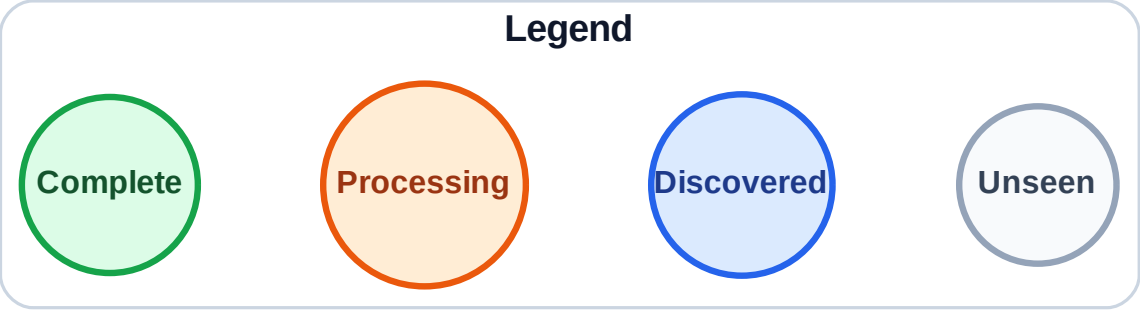
Stack (top at right): D | I | F | E | J

Visited: A, B, C

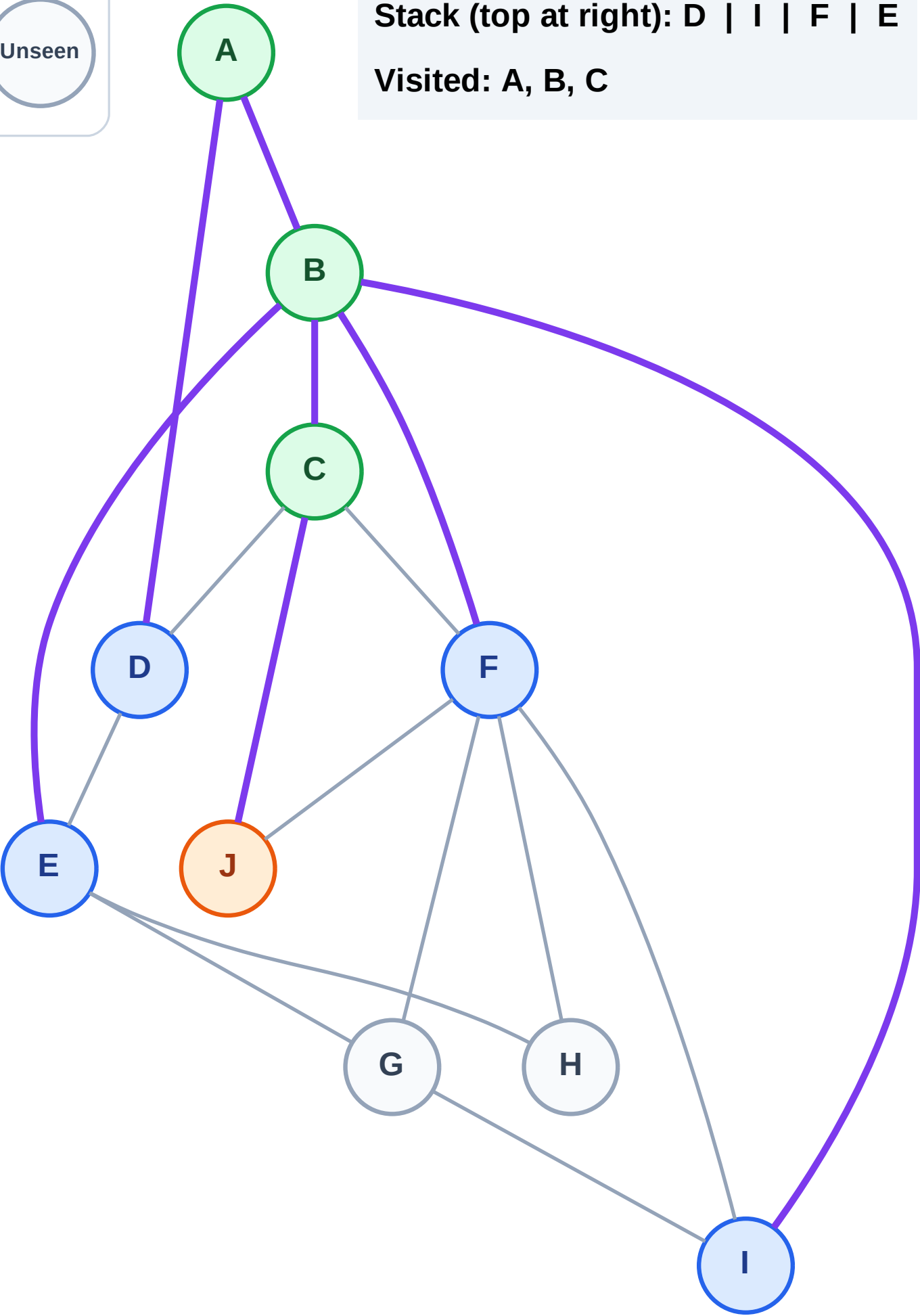


DFS Traversal

Step 8: Process node J



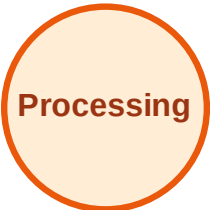
Stack (top at right): D | I | F | E
Visited: A, B, C



DFS Traversal

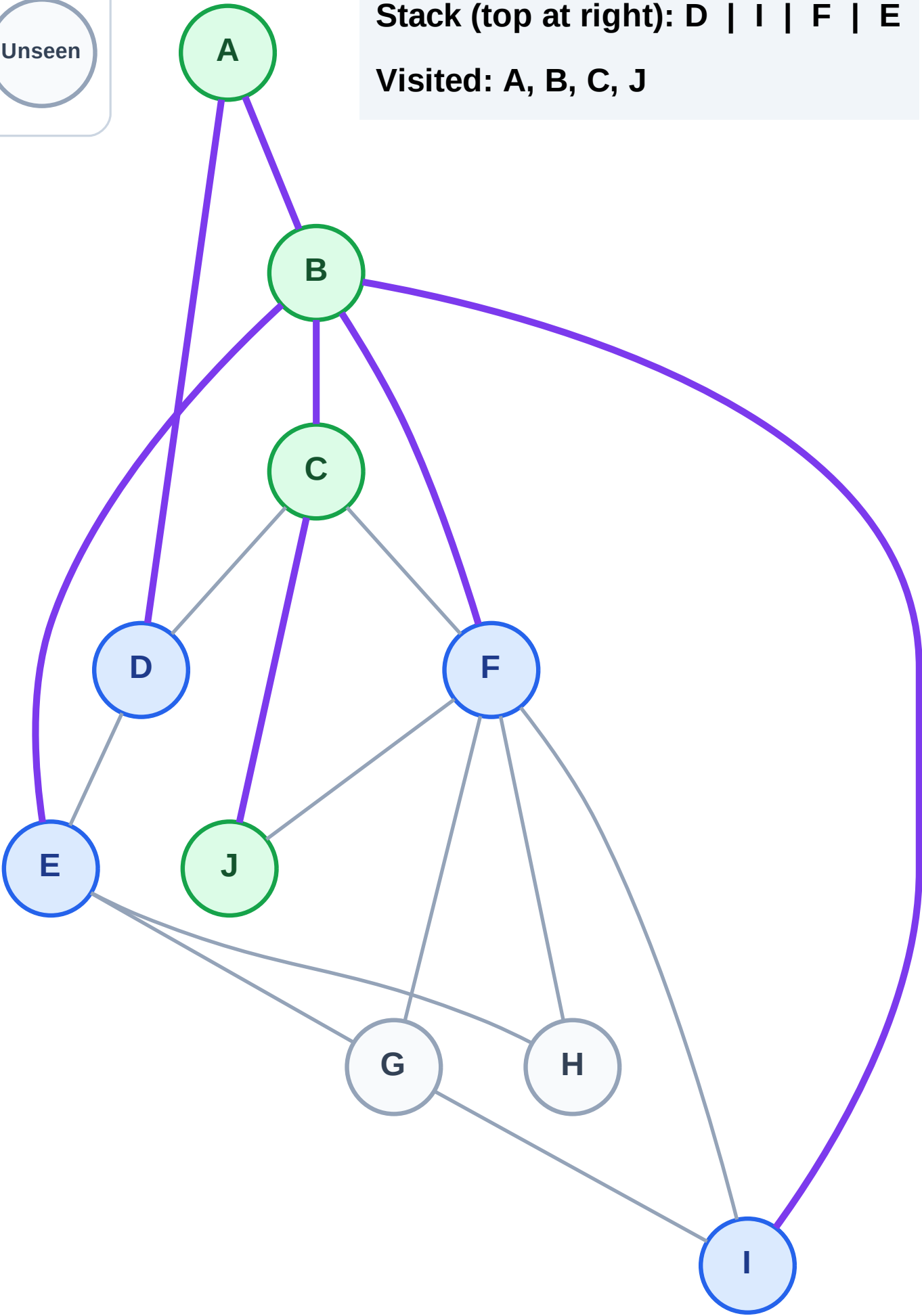
Step 9: No new neighbors from J

Legend



Stack (top at right): D | I | F | E

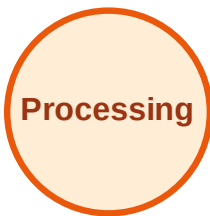
Visited: A, B, C, J



DFS Traversal

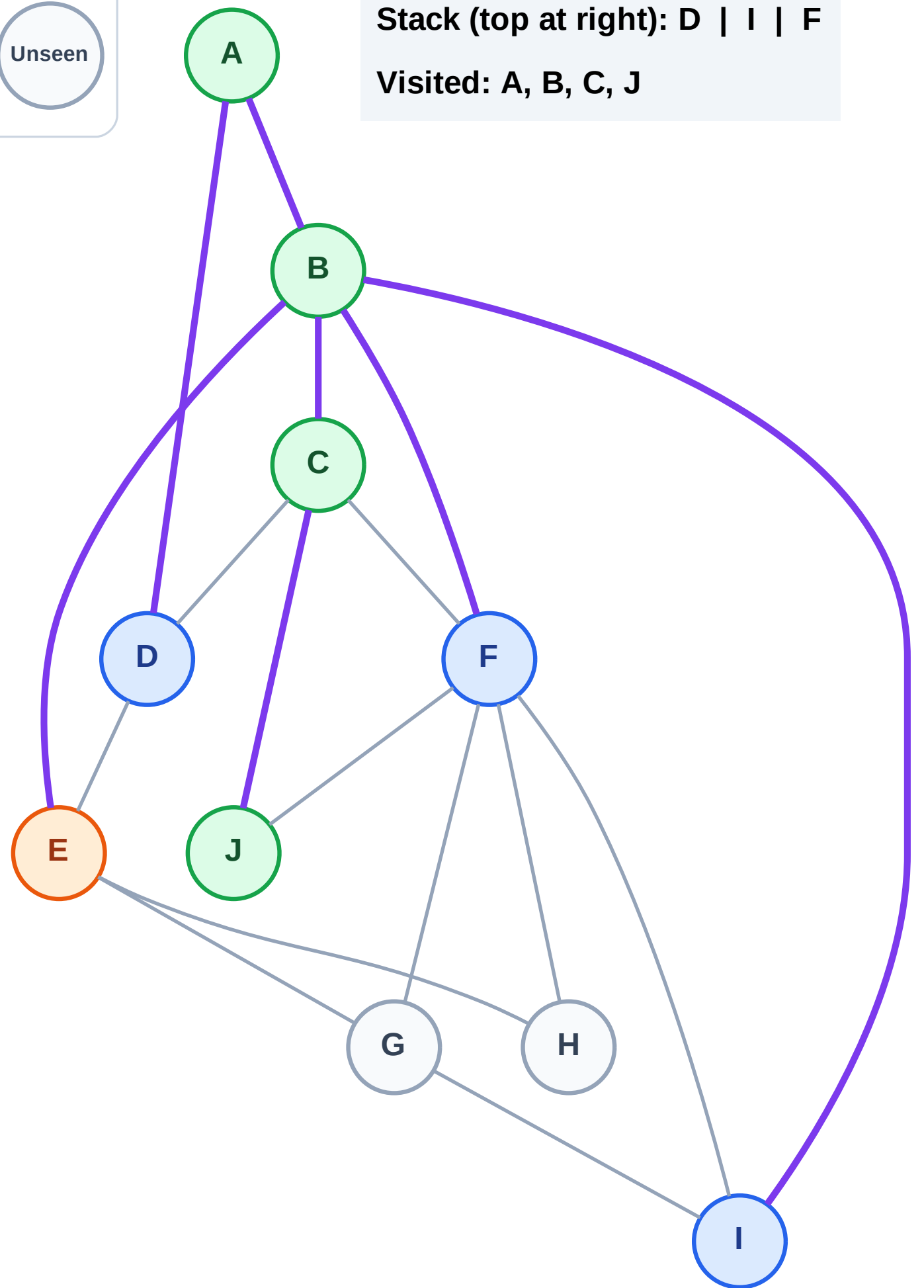
Step 10: Process node E

Legend



Stack (top at right): D | I | F

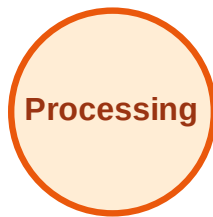
Visited: A, B, C, J



DFS Traversal

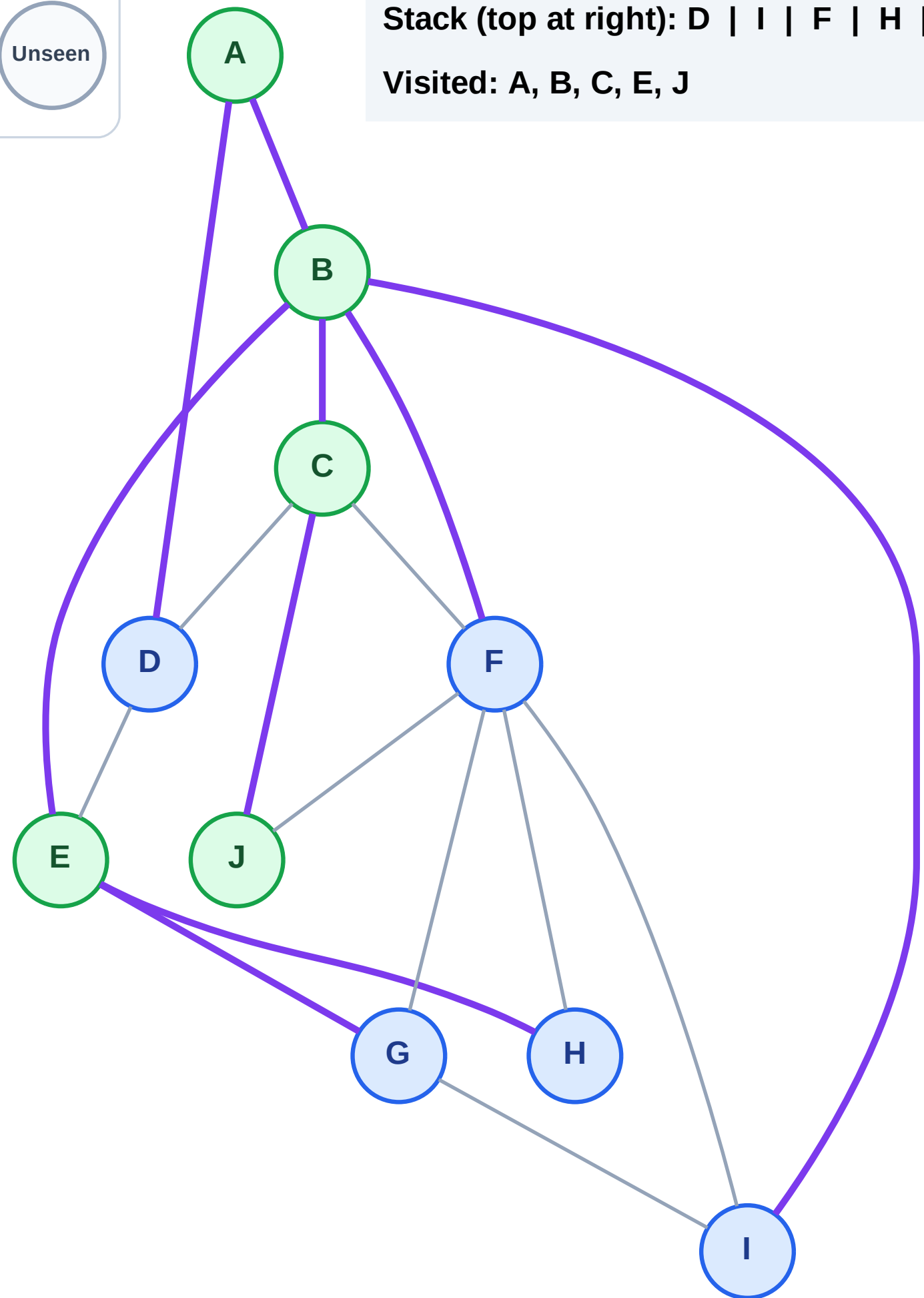
Step 11: Discover H, G from E

Legend

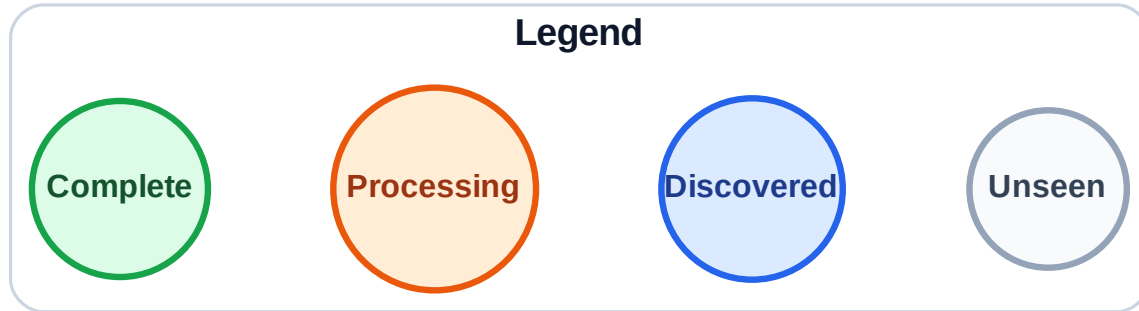


Stack (top at right): D | I | F | H | G

Visited: A, B, C, E, J

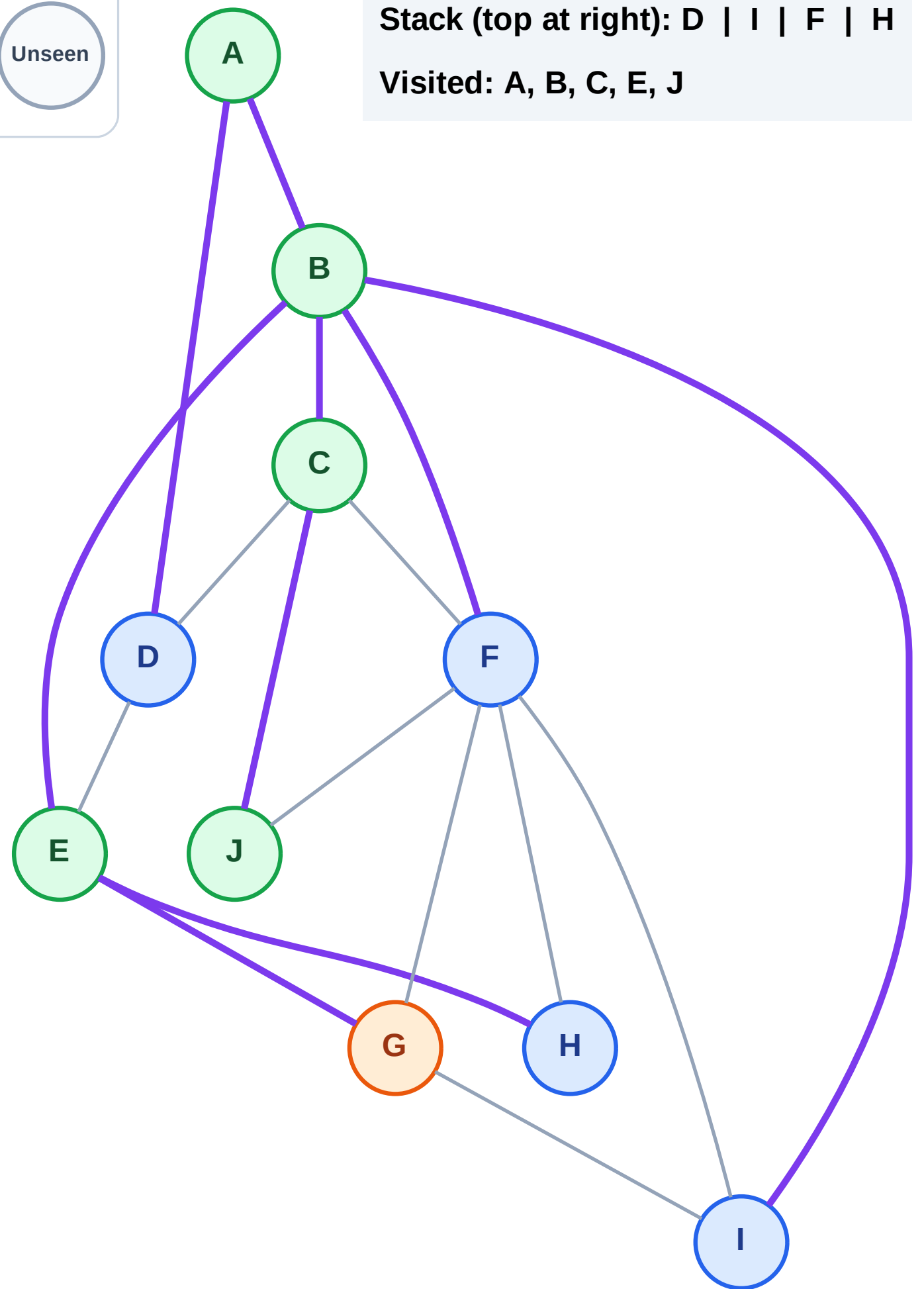


Step 12: Process node G



Stack (top at right): D | I | F | H

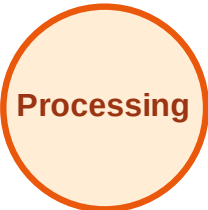
Visited: A, B, C, E, J



DFS Traversal

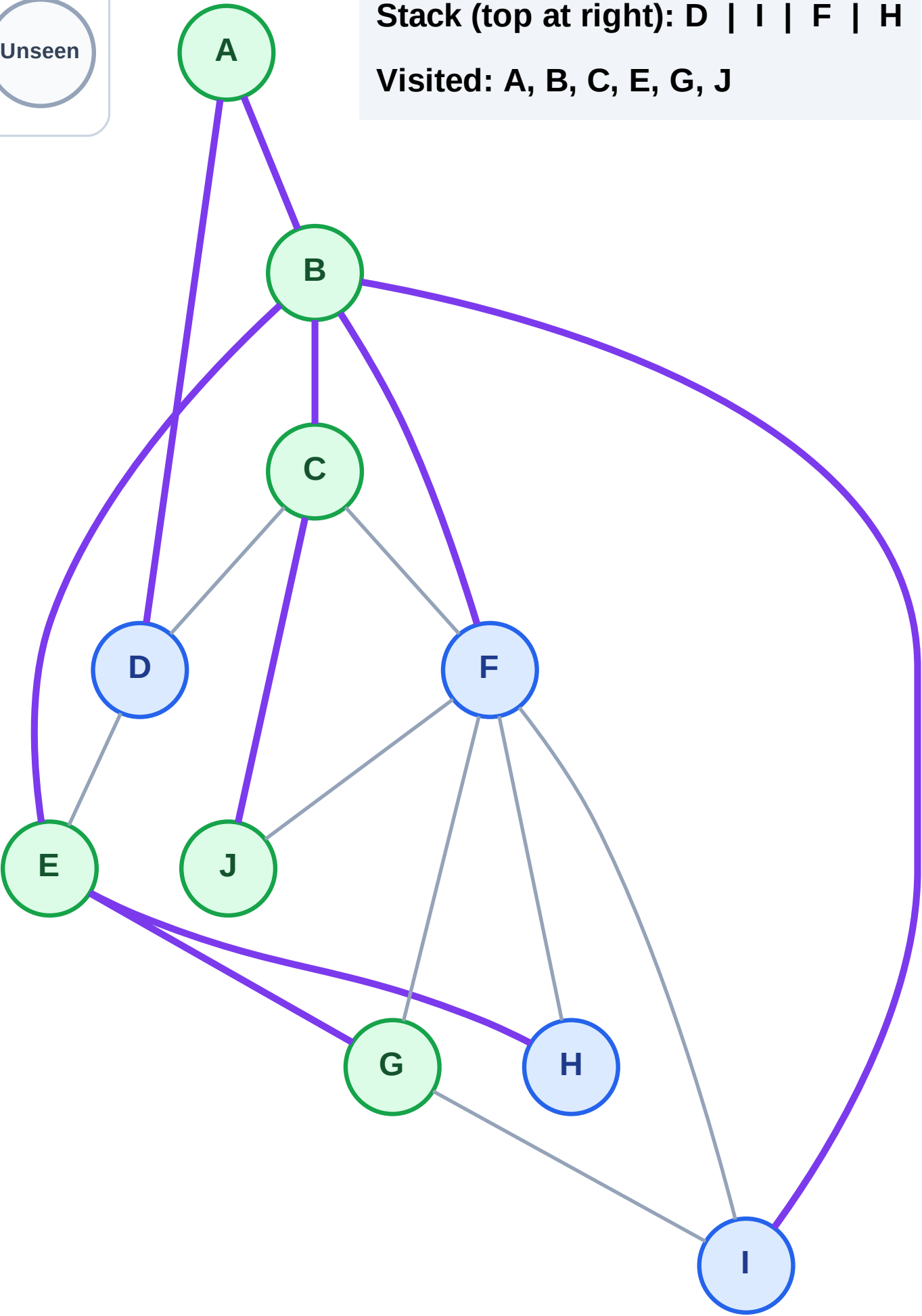
Step 13: No new neighbors from G

Legend



Stack (top at right): D | I | F | H

Visited: A, B, C, E, G, J



DFS Traversal

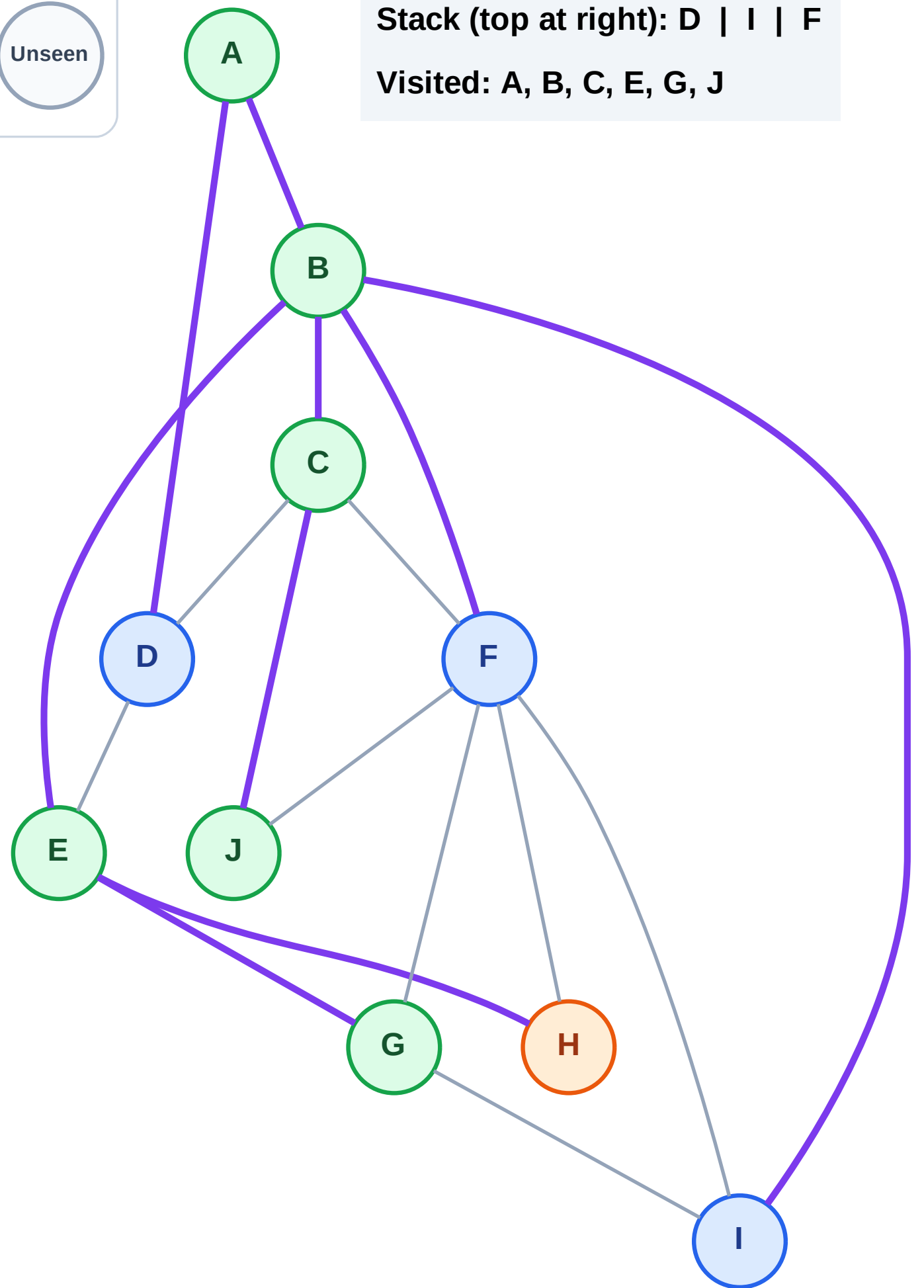
Step 14: Process node H

Legend



Stack (top at right): D | I | F

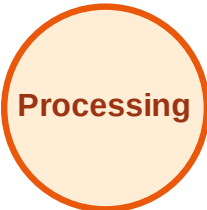
Visited: A, B, C, E, G, J



DFS Traversal

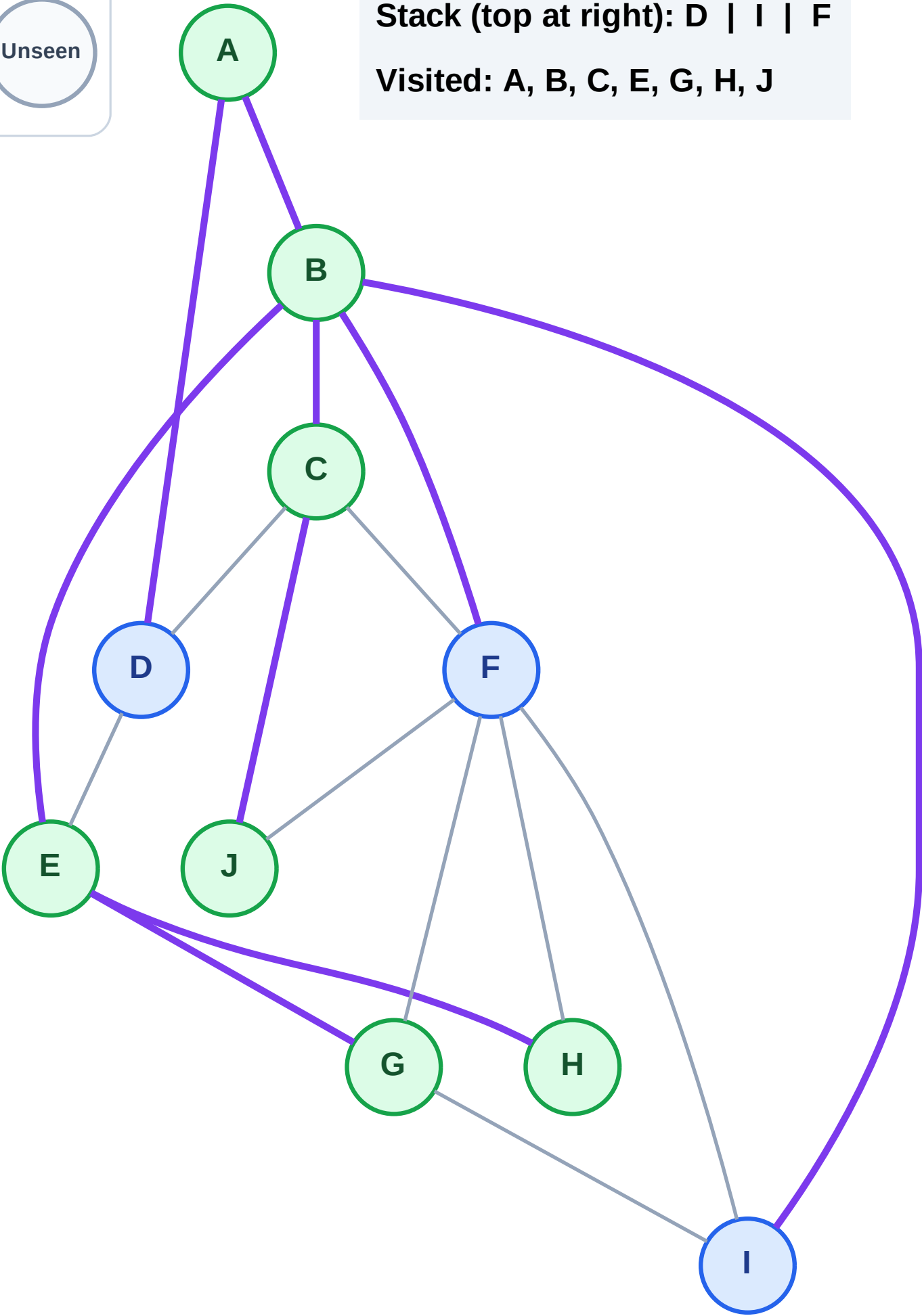
Step 15: No new neighbors from H

Legend



Stack (top at right): D | I | F

Visited: A, B, C, E, G, H, J



DFS Traversal

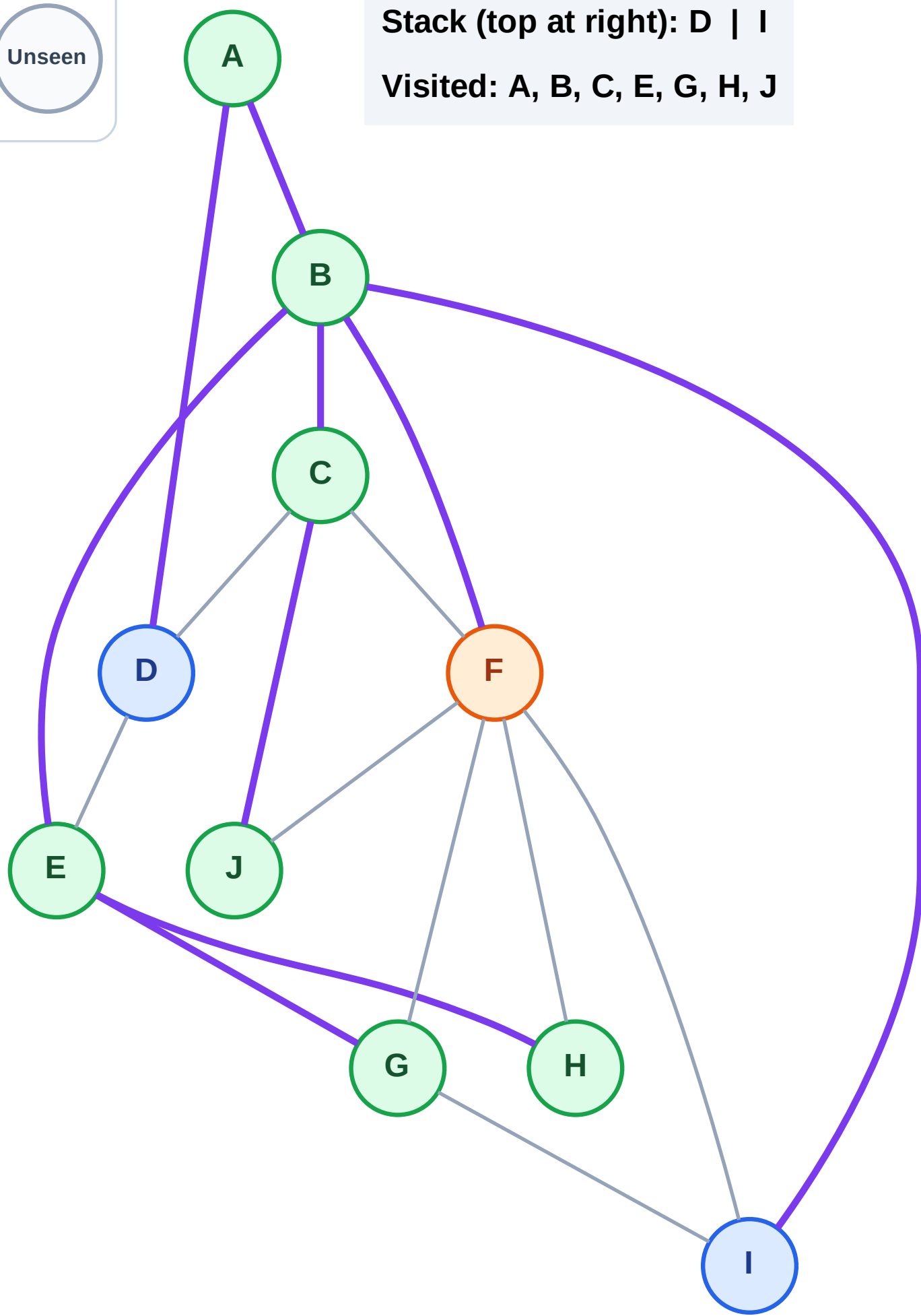
Step 16: Process node F

Legend



Stack (top at right): D | I

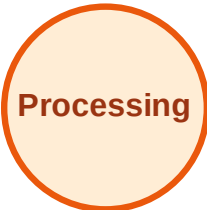
Visited: A, B, C, E, G, H, J



DFS Traversal

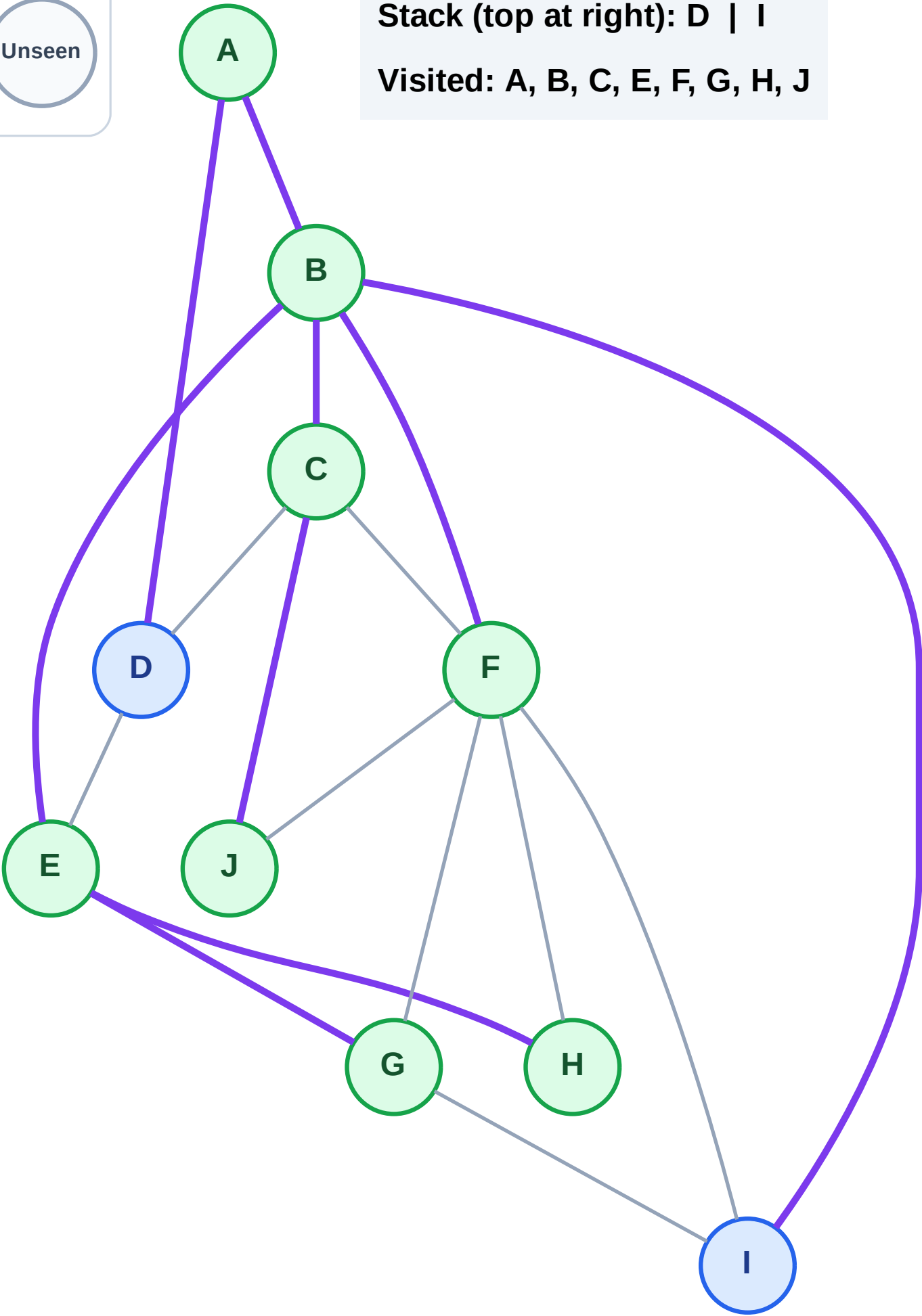
Step 17: No new neighbors from F

Legend



Stack (top at right): D | I

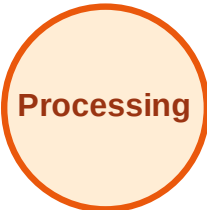
Visited: A, B, C, E, F, G, H, J



DFS Traversal

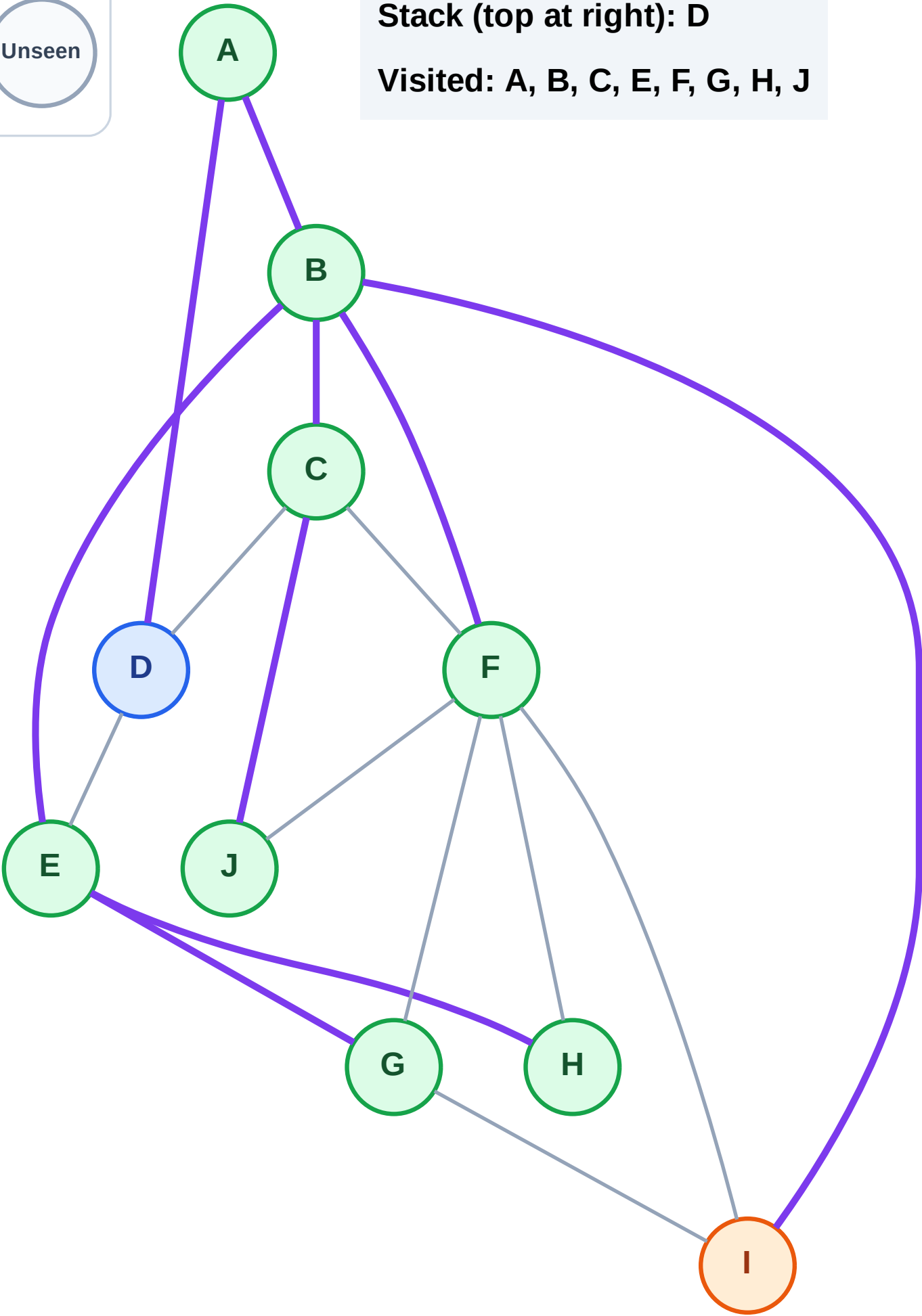
Step 18: Process node I

Legend



Stack (top at right): D

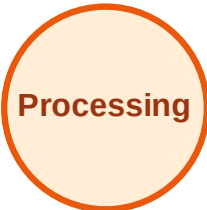
Visited: A, B, C, E, F, G, H, J



DFS Traversal

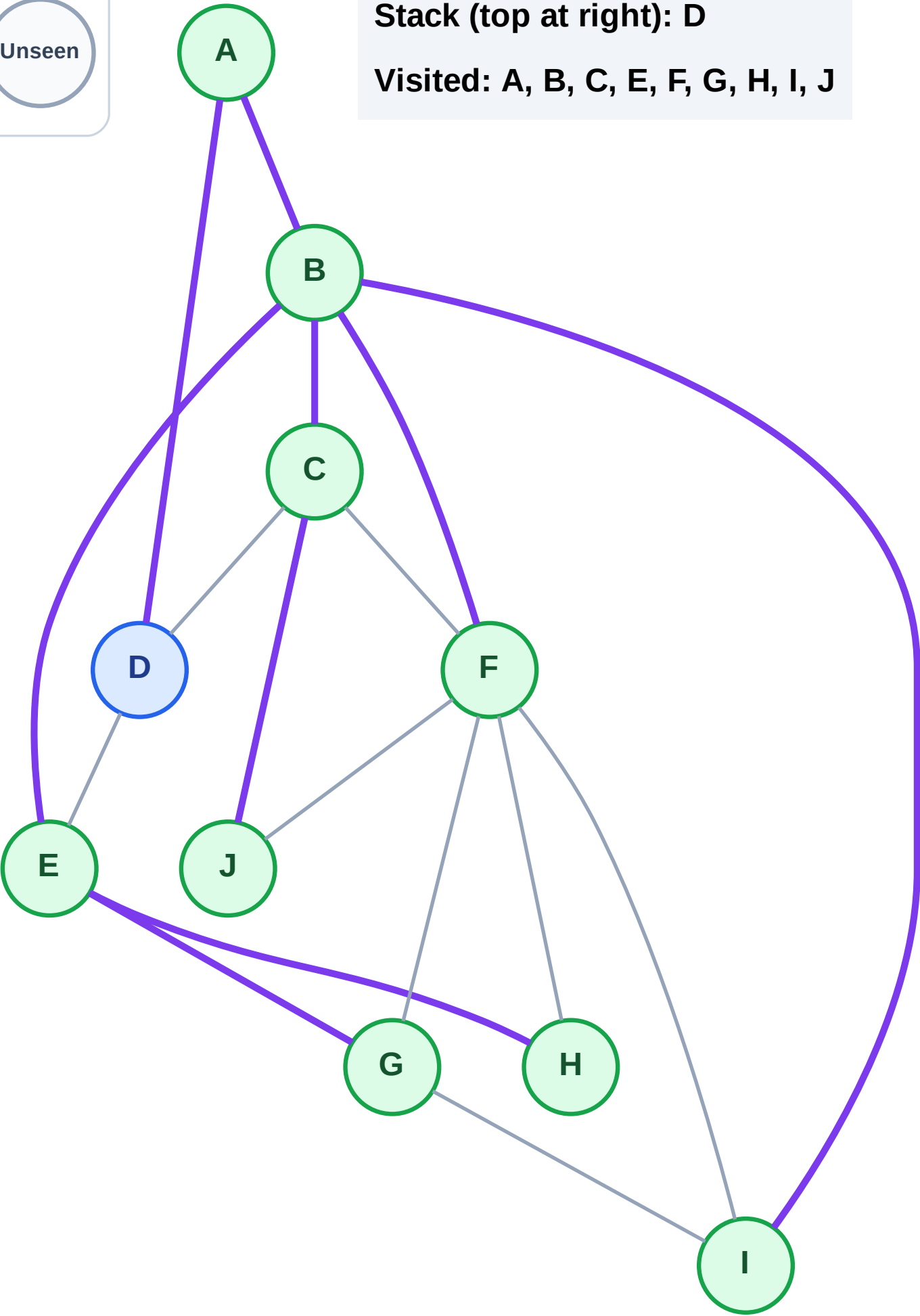
Step 19: No new neighbors from I

Legend



Stack (top at right): D

Visited: A, B, C, E, F, G, H, I, J



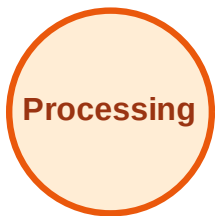
DFS Traversal

Step 20: Process node D

Legend



Complete



Processing



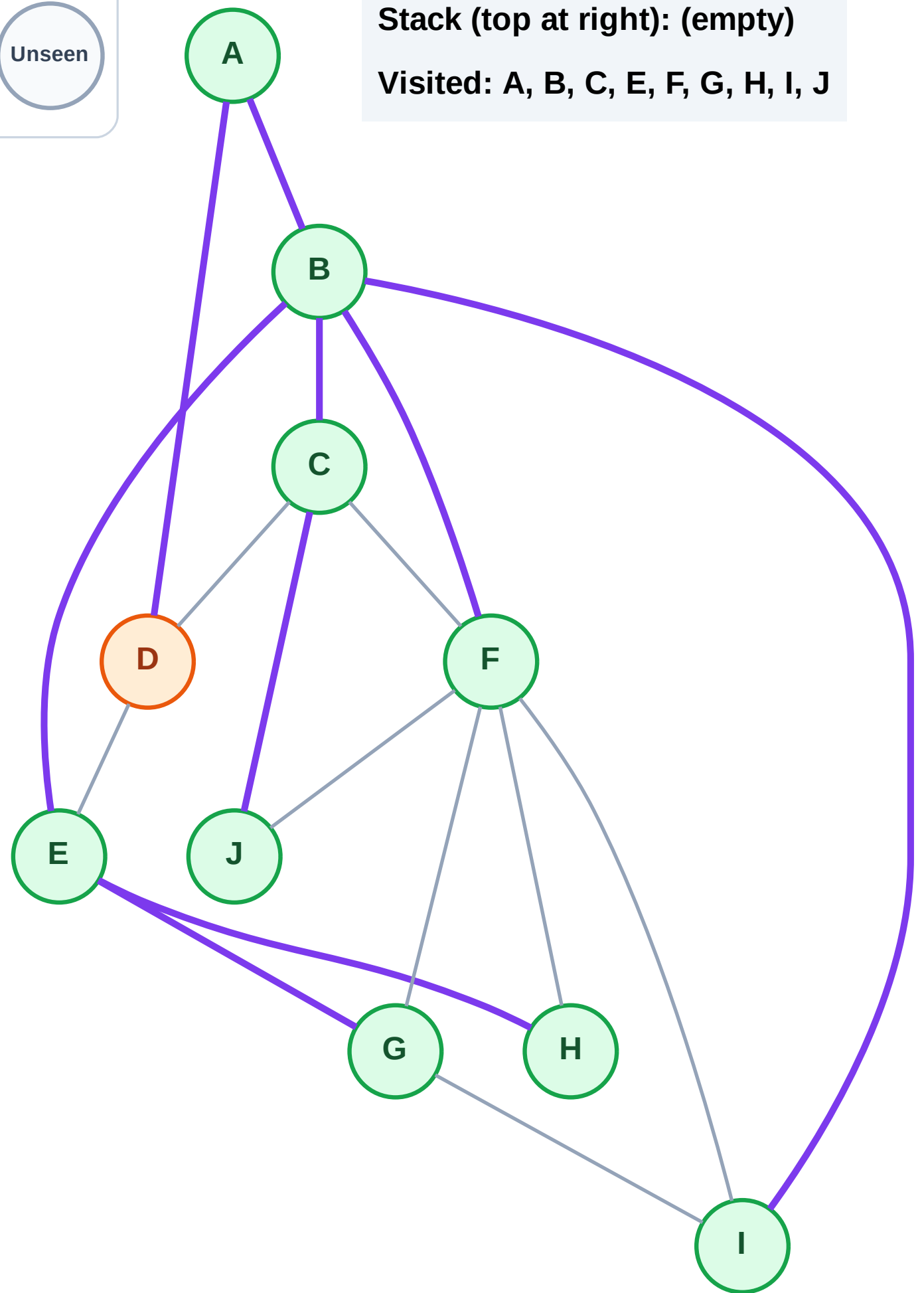
Discovered



Unseen

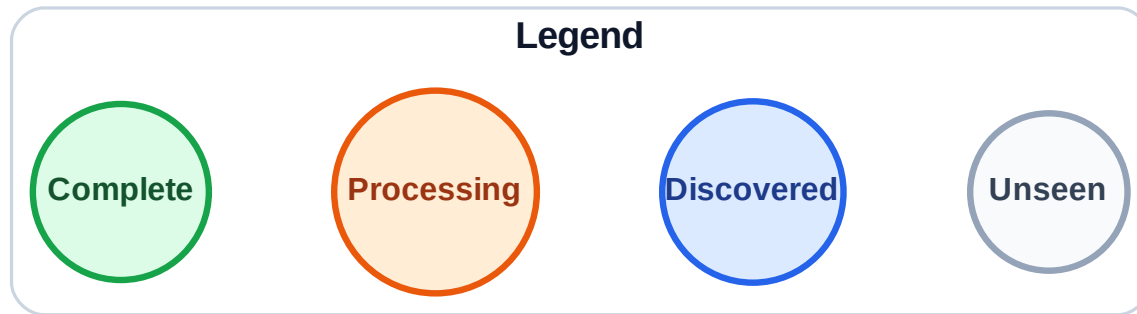
Stack (top at right): (empty)

Visited: A, B, C, E, F, G, H, I, J



Step 21: No new neighbors from D

Step 21: No new neighbors from D



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

